

MrktDB

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Problem Statement

Create a platform that aggregates all of the 13F data from SEC's website and provides insights into Hedge Fund holdings and associated trends by exhibiting visualizations based on the analysed data.

This should be a platform for alternative market data that would help investors generate Alpha. The platform should be able to provide access to multiple datasets that could be utilized by the users for generating inferences.

Process Model

Spiral Model

Spiral model is good for high risk or large projects where the requirements are ambiguous. The risks might be due to cost, schedule, performance, user interfaces, etc.

Risk analysis requires highly specific expertise, and a project's success is highly dependent on the risk analysis phase. It doesn't work well for smaller projects.

The most suitable model for this project would be the Spiral Model.

Justification:

The spiral model, also known as the spiral life cycle model, is a systems development life cycle model used in information technology (IT).
This model of development combines the features of the prototyping model and the waterfall model. The spiral model is favored for large, expensive, and complicated projects.

The steps in the spiral model iteration can be generalized as follows:

- **Customer Communication:** The system requirements are defined in as much detail as possible. This usually involves interviewing a number of users representing all the external or internal users and other aspects of the existing system.

- A preliminary design is created for the new system. This phase is the most important part of the "Spiral Model". In this phase all possible (and available) alternatives, which can help in developing a cost effective project are analyzed and strategies to use them are decided. This phase has been added specially in order to identify and resolve all the possible risks in the project development. If risks indicate any kind of uncertainty in requirements, prototyping may be used to proceed with the available data and find out possible solutions in order to deal with the potential changes in the requirements.

- A first prototype of the new system is constructed from the preliminary design. This is usually a scaled-down system, and Railway Reservation System represents an approximation of the characteristics of the final product.

- A second prototype is evolved by a fourfold procedure:
    - Evaluating the first prototype in terms of its strengths, weaknesses, and risks.
    - Defining the requirements of the second prototype
    - Planning and designing the second prototype
    - Constructing and testing the second prototype

MrktDB : General Project Description

Project Overview

A Platform for alternative market data and insights that helps retail investors[1] generate Alpha[2].

Alternative market data here is defined as anything other than the daily observed exchange trading data for listed securities. This data may be sourced from alternative sources and/or aggregated/analyzed to provide insights not easily accessible to retail investors.

Vision

The project (MrktDB) will be a platform for alternative market data and insights that would help investors generate Alpha. Evidently, the plan is to provide access to multiple datasets that could be utilized by the users. We will focus on aggregation and visualization of Hedge Fund Holdings using SEC (U.S. Securities and Exchange Commission) form 13F.

The **goal of the project** is to create a platform that aggregates all of the 13F data from SEC's website and provides insights into hedge fund holdings and associated trends by exhibiting **visualizations** based on the analysed data.

[1] **Retail Investors :** Retail investors are also called individual investors or retail traders. These are non-professional investors who purchase assets such as stocks, bonds, securities, mutual funds, and ETFs.

[2] **Alpha :** Alpha (α) is a term used in investing to describe an investment strategy's ability to beat the market, or it's "edge." Alpha is thus also often referred to as 'excess return'.

Software Requirements Specification

1. Introduction

1.1 Purpose

The purpose of this document is to present a detailed description of the 13F data visualization system (MrktDB). It will explain the purpose and features of the system, the interfaces of the system, what the system will do, the constraints under which it must operate and how the system will react to external stimuli. This document is intended for both the stakeholders and the developers of the system.

1.2 Scope

The Securities and Exchange Commission's (SEC) Form 13F is a quarterly report that is required to be filed by all institutional investment managers with **at least $100 million in assets under management**. Form 13F must be shared with the SEC within 45 days of the end of a calendar quarter. It requires the institutional investment managers that fit the criteria to disclose all of their long US listed positions and include instruments like common stock, ETFs, convertible debt securities, equity options, and warrants – a list that is approximately 17,500 instruments long! Short positions are not required to be disclosed and are not reported.

The regulation by its definition includes different types of institutional asset managers including, but not limited to, hedge funds, mutual funds, banks, trust companies, insurance companies, etc. ***This project will focus on the smartest people in the room – Hedge Funds.***

The number of hedge funds has had an exceptional growth curve in the last twenty years and has also been associated with several controversies.

Looking at some of the search engine results for associated keywords, *there are hundreds of thousands of google searches for 'hedge fund' or related keywords every month in the United States alone.*

The Hedge Funds managers have a reputation of consistently beating the market and producing outsized returns. Hedge Funds being the biggest players in the industry carry out market research on a big level. Stock brokers and people who trade in the market also tend to research a lot before holding any positions.

Hedge Fund investments are one of the primary sources to get research data for normal users, but it isn't so easy to gather it all.

MrktDB will provide its users insights into the Hedge Fund holdings by **aggregating** and **visualizing** all 13F data! The charts on the website will display historic market data and trends in the holdings of all 13F filers.

1.3 Motivation

Wolves are *intelligent*, *fearless*, and *ferocious*. They are always on the prowl and jump upon the first opportunity they get. Sometimes they hunt alone while at other times they create a wolf pack and bring the prize home. In a wolf pack they have *Alpha*, *Beta*, and *Gamma* wolves – setting a social hierarchy based on one's capabilities.

Similarly, Hedge Fund managers are *smart*, *sophisticated*, *and highly opportunistic*. They have few rules that they play with and they have ruled the financial markets for a long time. They have been known to create wolf packs to bring down their prey. The Alpha wolves have created tons of financial Alpha for the investors who have put their faith in them over the years. *MrktDB will provide a sneak peek into the minds of these wolves*.

These hedge fund managers have a reputation of consistently beating the market and producing outsized returns. Some of the well-known Hedge Funds are Bridgewater Associates, Scion Capital, Tiger Capital, Oaktree Capital, Renaissance Capital, etc. What these sophisticated investors are buying or selling is of great interest to the market and it is not unusual to see news of a prominent hedge fund buying or selling a stock and associated big market movements in that stock. For example, if a typical Hedge Fund thinks something is worth buying, it must surely be!

MrktDB will provide its users insights into the Hedge Fund holdings by **aggregating** and **visualizing** data under the following broad categories:

1. Individual Fund Dashboard
2. Individual Stock[3] Profile
3. Aggregated Funds Dashboard
4. Personalized User Dashboard

[3]**Stock :** A stock is a security that represents the ownership of a fraction of a corporation. This entitles the owner of the stock to a proportion of the corporation's assets and profits equal to how much stock they own. Units of stock are called "shares."

1.4 Definitions, Acronyms, and Abbreviations

1.4.1 Abbreviations

SEC - Securities and Exchange Commission

API - Application Programming Interface

QoQ - Quarter on Quarter

UI - User Interface

DFD - Data Flow Diagram

PaaS - Platform as a Service

AWS - Amazon Web Services

SSH - Secure Shell

JSON - JavaScript Object Notation

EDGAR - Electronic Data Gathering, Analysis and Retrieval

1.4.2 Definitions

Securities and Exchange Commission (SEC) : The U.S. Securities and Exchange Commission (SEC) is an independent federal government regulatory agency responsible for protecting investors, maintaining fair and orderly functioning of the securities markets, and facilitating capital formation. The SEC promotes full public disclosure, protects investors against fraudulent and manipulative practices in the market, and monitors corporate takeover actions in the United States.

S&P 500 : The S&P 500 Index or the Standard & Poor's 500 Index is a market-capitalization-weighted index of the 500 largest U.S. publicly traded companies.

Form 13F : The Securities and Exchange Commission's (SEC) Form 13F is a *quarterly report* that is required to be filed by all institutional investment managers with at least $100 million in assets under management. Form 13F must be shared with the SEC *within 45 days* of the end of a calendar quarter.

Edgar Database : EDGAR — Electronic Data Gathering, Analysis and Retrieval — is the electronic filing system created by the Securities and Exchange Commission to increase the efficiency and accessibility of corporate filings. The system is used by all publicly traded companies when submitting required documents to the SEC.

Prices: Open, High, Low, Close : In stock trading, the **high** and **low** refer to the maximum and minimum prices in a given time period. **Open** and **close** are the prices at which a stock began and ended trading in the same period. **Volume** is the total amount of trading activity.

Hedge funds : They're only open to "accredited" or qualified investors: Hedge funds are only allowed to take money from "qualified" investors—individuals with an annual income that exceeds $200,000 for the past two years or a net worth exceeding $1 million, excluding their primary residence.

They offer wider investment latitude than other funds: A hedge fund's investment universe is only limited by its mandate. A hedge fund can basically invest in anything—land, real estate, stocks, derivatives, and currencies. Mutual funds, by contrast, have to basically stick to stocks or bonds and are usually long-only.

They often employ leverage: Hedge funds often use borrowed money to amplify their returns.

Fee structure: Instead of charging an expense ratio only, hedge funds charge both an expense ratio and a performance fee. This fee structure is known as “Two and Twenty” — a 2% asset management fee and then a 20% cut of any gains generated.

1.5 Overview of the document

The rest of the document deals with all the main features of this application. It not only describes various functions but also gives details about how these functions are related to each other. Apart from the data flow diagrams, the document also contains cost estimates for developing this system. Various risks associated with the system have also been mentioned along with the ways to mitigate them.

The next chapter, the Overall Description section, of this document gives an overview of the functionality of the product. It describes the informal requirements and is used to establish a context for the technical requirements specification in the next chapter.

The third chapter, Requirements Specification section, of this document is written primarily for the developers and describes in technical terms the details of the functionality of the product.

Both sections of the document describe the same software product in its entirety, but are intended for different audiences and thus use different language.

2. The Overall Description

2.1 Product Perspective

MrktDB is a product that requires some additional hardware and software interfaces to function which includes the OS, a web browser, and a stable internet connection. When released, the final product would be the first version of the application. It is designed as a user centered product, which could be accessed to give a personalised experience to any authenticated user. The system restricts access to some of its components to users who try to access the website without authentication.

2.1.1 External System Interfaces

There are a few external interfaces being used in this project-

- ➔ Data being scraped from EDGAR will have to be stored in a 13F database.
- ➔ User data is stored in the User database.
- ➔ Some elements on the website will be embedded from 3rd party tools like TradingView and FontAwesome.

2.1.2 Hardware Interfaces

2.1.2.1 Frontend

There is no specified screen resolution for accessing the website. The website can be used on any screen size.

2.1.2.2 Backend

In this project, we will leverage the use of cloud computing on AWS. The application will be deployed on AWS Elastic Beanstalk and which is a PaaS.

The application will be served by a t2.micro instance which provides us with- **1 vCPU**, **2 GB RAM**, and **8 GB storage**. Storage capacity will have to be increased upto 50GB to store all required data. The database will also be hosted on another similar EC2 instance. All server-side computations are done on these instances. Both instances can be accessed remotely via SSH.

The user can access the website on any device through a web browser.

2.1.3 Software Interfaces

The software interface will follow the Model-View-Controller (MVC) model for rendering and modeling data objects. Data in the database will be stored in JSON format.

The application will be implemented in Django and it will use React for the front-end. VS Code will be used as a text editor while implementing. MongoDB will be used as DBMS. Git and GitHub will be used for version control.

The softwares required in the development of the application include-

- Django as a web framework
- React JS for frontend
- VS Code as a text editor
- Git and GitHub for version control
- HTML
- CSS
- SASS
- MongoDB as a database
- Chart.js as a charting library
- Python will be used for writing scripts to scrape data
- Selenium
- Beautiful Soup

2.1.4 UI Interface

- The product website will be made very simplistic and responsive, so it will be easy for the users to access. It will be able to run on any device.
- Though the UI will be developed to look a little bit different on desktop and mobile devices, most of the functionalities will be available to both types of users.
- Content on extra-large screens will be displayed the same as on desktop as a max-width of 1500px will be set for the website contents.
- The main attraction of the website will be the visualizations that will be displayed. Users will be able to look at historic trends and analyse market data that is not easily accessible.

- **Required web pages include-**

    - **Signup/Signin pages:** Forms for login and signup for users.
    - **Additional details form:** Form to obtain additional details of a user.
    - **Personalised user dashboard:** User dashboard to display watchlisted funds and stocks.
    - **Individual Fund page:** Separate page that displays data, tables, and charts about particular funds.
    - **Individual Stock page:** Separate page that displays data, tables, and charts about particular stocks.
    - **Billing Portal:** Payment page for access to premium features.
    - **Help/Suggestions page:** Page through which users can give their opinions and leave any comments.
    - **FAQ page:** A page to display answers to general questions that the users might have about the website.

2.1.5 Communication Interface

The system requires HTTP to communicate with the database. The system and database can be configured to be accessed via any available port.

The web based UI is the only means of communication between the user and the system, though the user can directly send an email to the admin in case of any queries. The system is accessible through all popular web browsers that interact with HTML pages.

Dedicated email address so that the user can communicate with the admin.

2.1.6 Operations

- Users can analyse historical performance of Hedge Funds and other filers by looking at the charts on the website.
- Users can also mark their favourite funds and stocks to display on the user dashboard.
- Every night at 2 a.m, a script will run to check for any new data and store it to the database.

2.2 Product Functions

The application will be capable of performing the following functions. The functions depend on the user's level and permission package, as explained in the user characteristics.

- Provide each user with an account.
- Users can see and change their personal information after verifying new information.
- Application would use cookies to store temporary session data.
- Search functionality so that users can easily search for the fund/stock they want to see.
- Users must be able to put their favourite funds/stocks in a watchlist that will be displayed on the user dashboard for easy access.
- Various types of charts and tables would be displayed on the website to understand the trends in the market.
- FAQ page for the users to clarify certain doubts about the website.
- Users can contact the admin for any particular queries.
- Functionality to print and download 13F reports.

2.3 User Characteristics

The intended users of this application will be people who are inclined towards finance and the stock market sector. Other people who perform trading on a regular basis can also benefit from the application.

The users are expected to be Internet literate and be able to interpret charts and tables.

2.4 Constraints

- The application will only be available in english.
- There wouldn't be any hardware limitations as the website will be deployed on AWS and would be self-scaling.
- For the initial part of the project, 13F data from 2013 to the present time will be scraped. We can't access the data filed before 2013 at this moment as it is in .txt format, though ways are still being devised for the same.
- Companies do not file 13F data on a fixed time in a given quarter. The data failed to file in the appropriate time is called 'Fails to deliver data'. So, a script will have to be kept running that will update the database every night.
- SEC's current guidelines limit users to a total of no more than 10 requests/sec.

2.5 Assumptions and Dependencies

- A web browser is required to access the website.
- Users must be connected to the Internet.
- Users must know how to read charts.
- Basic knowledge of the stock market.
- There are some dependencies on 3rd party softwares like TradingView and Font Awesome.

2.6 Apportioning of Requirements

- Version 1 of the application is available only in english.
- Most popular funds and stocks charts on MrktDB have to be delayed until further versions due to the lack of a user community in the initial few weeks.
- Functionality to download 13F data in excel format.

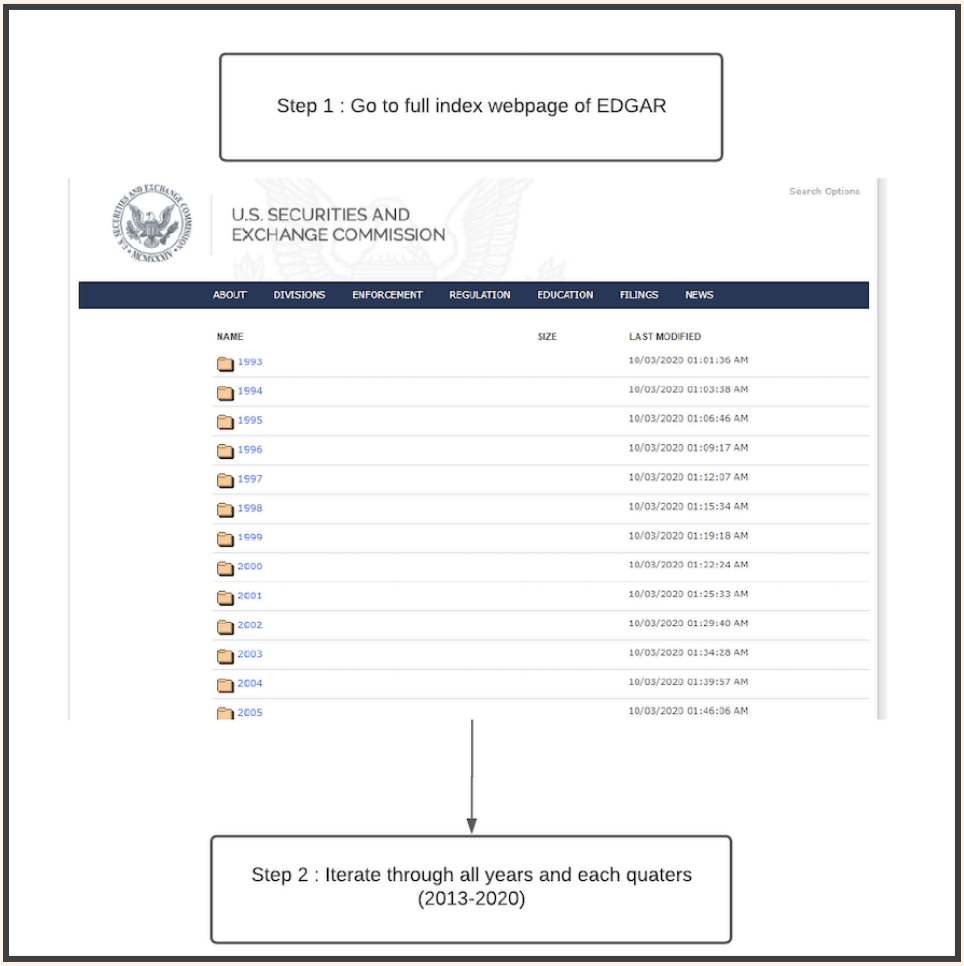
3. Specific Requirements

3.1 Functional Requirements

3.1.1 Get CIK List (13F-HR)

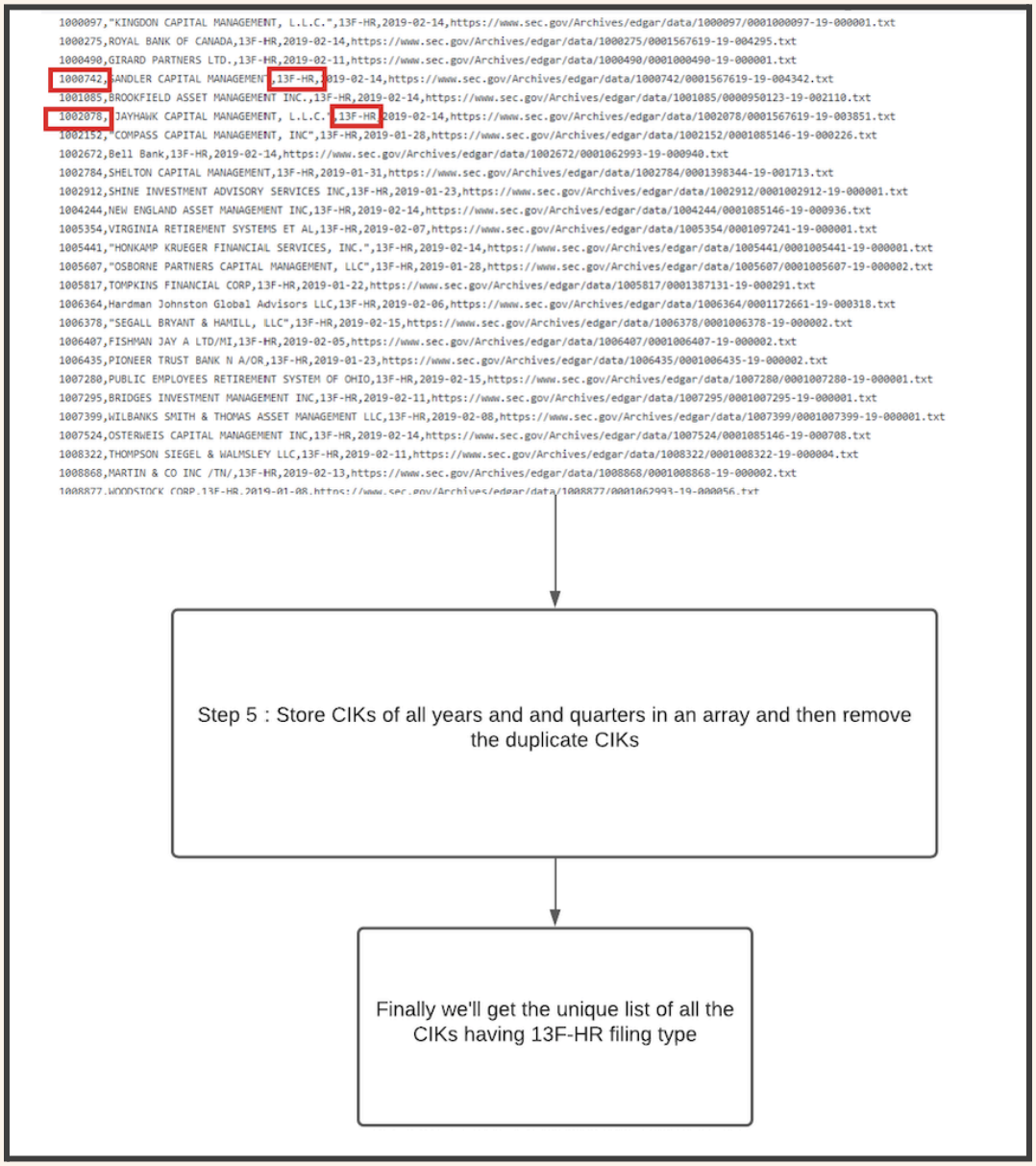
Company CIKs are needed to scrape data from EDGAR. The scraping script will retrieve data based on the CIKs feeded to it.

The process is shown in the diagram below-



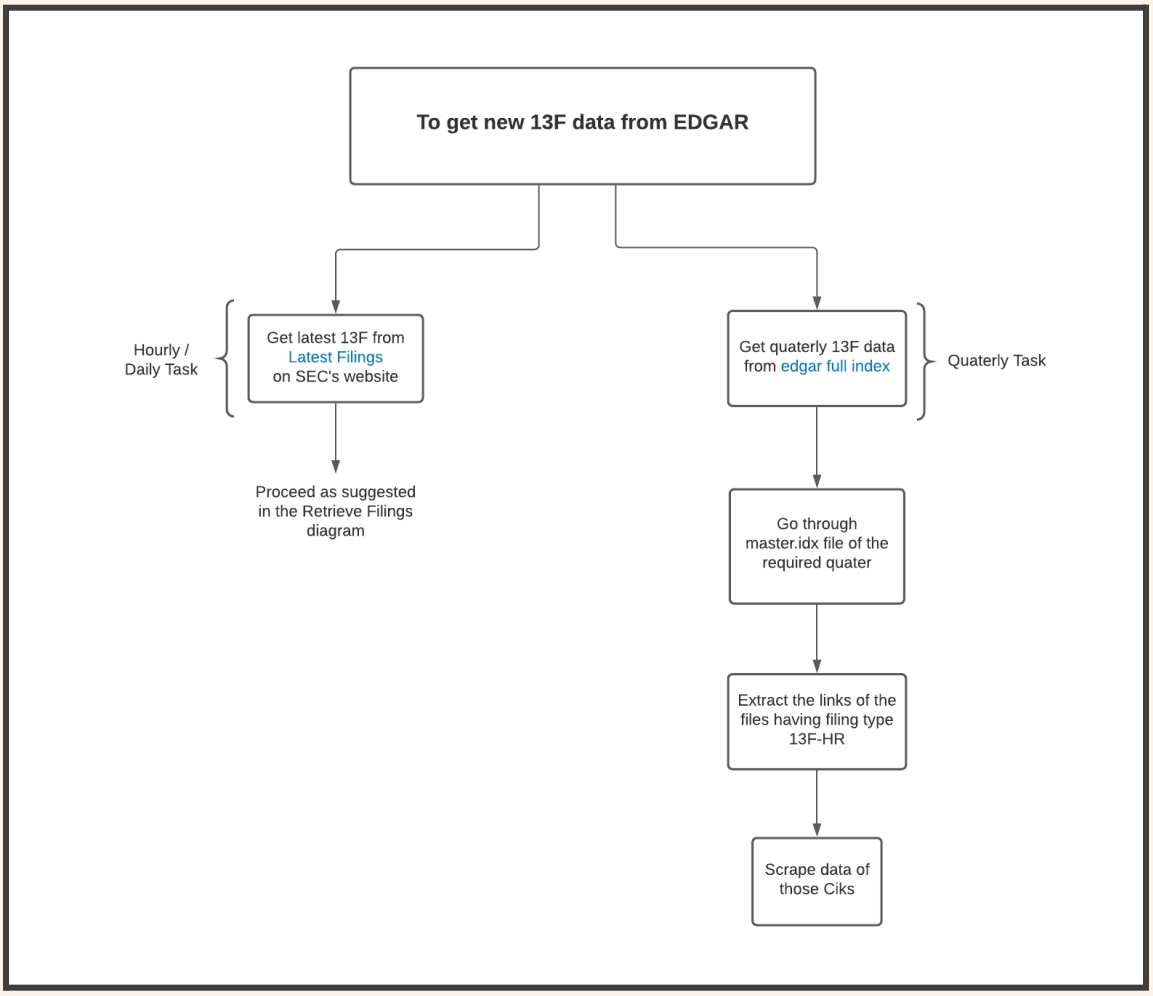

U.S. SECURITIES AND EXCHANGE COMMISSION
Search Options
ABOUT
DIVISIONS
ENFORCEMENT
REGULATION
EDUCATION
FILINGS
NEWS
Parent Directory
NAME
SIZE
LAST MODIFIED
QTR1
09/26/2020 03:21:49 AM
QTR2
09/26/2020 03:23:26 AM
QTR3
09/30/2020 10:24:57 PM
QTR4
10/01/2020 12:20:20 AM
Site Map | Accessibility | Contracts | Privacy | Inspector General | Careers | Contact
FOIA | No FEAR Act & EEO Data | Open Government | Plain Writing | Links | Investor.gov | USA.gov
Step 3 : Go to master.idx file of each quarter which have all the data of that particular quarter.


company.zip 3927 KB 03/31/2020 10:14:00 PM
crawler.idx 57781 KB 03/31/2020 10:09:54 PM
form.gz 4831 KB 03/31/2020 10:14:18 PM
form.idx 47162 KB 03/31/2020 10:10:24 PM
form.Z 7979 KB 03/31/2020 10:14:10 PM
form.zip 4465 KB 03/31/2020 10:14:48 PM
master.gz 3821 KB 03/31/2020 10:15:02 PM
master.idx 27988 KB 03/31/2020 10:09:39 PM
master.Z 7040 KB 03/31/2020 10:14:57 PM
master.zip 3636 KB 03/31/2020 10:15:26 PM
sitemap.quarterlyindex1.xml 6406 KB 03/31/2020 10:09:17 PM
Step 4 : Extract rows having 13F filing type
From here we get ciks of the filings


1000097,"KINGDON CAPITAL MANAGEMENT, L.L.C.",13F-HR,2019-02-14,https://www.sec.gov/Archives/edgar/data/1000097/0001000097-19-000001.txt
1000275,ROYAL BANK OF CANADA,13F-HR,2019-02-14,https://www.sec.gov/Archives/edgar/data/1000275/0001567619-19-004295.txt
1000490,GIRARD PARTNERS LTD.,13F-HR,2019-02-11,https://www.sec.gov/Archives/edgar/data/1000490/0001000490-19-000001.txt
1000742,CANDLER CAPITAL MANAGEMENT,13F-HR,2019-02-14,https://www.sec.gov/Archives/edgar/data/1000742/0001567619-19-004342.txt
1001085,BROOKFIELD ASSET MANAGEMENT INC.,13F-HR,2019-02-14,https://www.sec.gov/Archives/edgar/data/1001085/0000950123-19-002110.txt
1002078,JAYHAWK CAPITAL MANAGEMENT, L.L.C.,13F-HR,2019-02-14,https://www.sec.gov/Archives/edgar/data/1002078/0001567619-19-003851.txt
1002152,"COMPASS CAPITAL MANAGEMENT, INC",13F-HR,2019-01-28,https://www.sec.gov/Archives/edgar/data/1002152/0001085146-19-000226.txt
1002672,Bell Bank,13F-HR,2019-02-14,https://www.sec.gov/Archives/edgar/data/1002672/0001062993-19-000940.txt
1002784,SHELTON CAPITAL MANAGEMENT,13F-HR,2019-01-31,https://www.sec.gov/Archives/edgar/data/1002784/0001398344-19-001713.txt
1002912,SHINE INVESTMENT ADVISORY SERVICES INC,13F-HR,2019-01-23,https://www.sec.gov/Archives/edgar/data/1002912/0001002912-19-000001.txt
1004244,NEW ENGLAND ASSET MANAGEMENT INC,13F-HR,2019-02-14,https://www.sec.gov/Archives/edgar/data/1004244/0001085146-19-000936.txt
1005354,VIRGINIA RETIREMENT SYSTEMS ET AL,13F-HR,2019-02-07,https://www.sec.gov/Archives/edgar/data/1005354/0001097241-19-000001.txt
1005441,"HONKAMP KRUEGER FINANCIAL SERVICES, INC.",13F-HR,2019-02-14,https://www.sec.gov/Archives/edgar/data/1005441/0001005441-19-000001.txt
1005607,"OSBORNE PARTNERS CAPITAL MANAGEMENT, LLC",13F-HR,2019-01-28,https://www.sec.gov/Archives/edgar/data/1005607/0001005607-19-000002.txt
1005817,TOMPKINS FINANCIAL CORP,13F-HR,2019-01-22,https://www.sec.gov/Archives/edgar/data/1005817/0001387131-19-000291.txt
1006364,Hardman Johnston Global Advisors LLC,13F-HR,2019-02-06,https://www.sec.gov/Archives/edgar/data/1006364/0001172661-19-000318.txt
1006378,"SEGALL BRYANT & HAMILL, LLC",13F-HR,2019-02-15,https://www.sec.gov/Archives/edgar/data/1006378/0001006378-19-000002.txt
1006407,FISHMAN JAY A LTD/MI,13F-HR,2019-02-05,https://www.sec.gov/Archives/edgar/data/1006407/0001006407-19-000002.txt
1006435,PIONEER TRUST BANK N A/OR,13F-HR,2019-01-23,https://www.sec.gov/Archives/edgar/data/1006435/0001006435-19-000002.txt
1007280,PUBLIC EMPLOYEES RETIREMENT SYSTEM OF OHIO,13F-HR,2019-02-15,https://www.sec.gov/Archives/edgar/data/1007280/0001007280-19-000001.txt
1007295,BRIDGES INVESTMENT MANAGEMENT INC,13F-HR,2019-02-11,https://www.sec.gov/Archives/edgar/data/1007295/0001007295-19-000001.txt
1007399,WILBANKS SMITH & THOMAS ASSET MANAGEMENT LLC,13F-HR,2019-02-08,https://www.sec.gov/Archives/edgar/data/1007399/0001007399-19-000001.txt
1007524,OSTERWEIS CAPITAL MANAGEMENT INC,13F-HR,2019-02-14,https://www.sec.gov/Archives/edgar/data/1007524/0001085146-19-000708.txt
1008322,THOMPSON SIEGEL & WALMSLEY LLC,13F-HR,2019-02-11,https://www.sec.gov/Archives/edgar/data/1008322/0001008322-19-000004.txt
1008868,MARTIN & CO INC /TN/,13F-HR,2019-02-13,https://www.sec.gov/Archives/edgar/data/1008868/0001008868-19-000002.txt
1008877,WOODSTOCK CORP,13F-HR,2019-01-08,https://www.sec.gov/Archives/edgar/data/1008877/0001062993-19-000056.txt
Step 5 : Store CIKs of all years and and quarters in an array and then remove the duplicate CIKs
Finally we'll get the unique list of all the CIKs having 13F-HR filing type

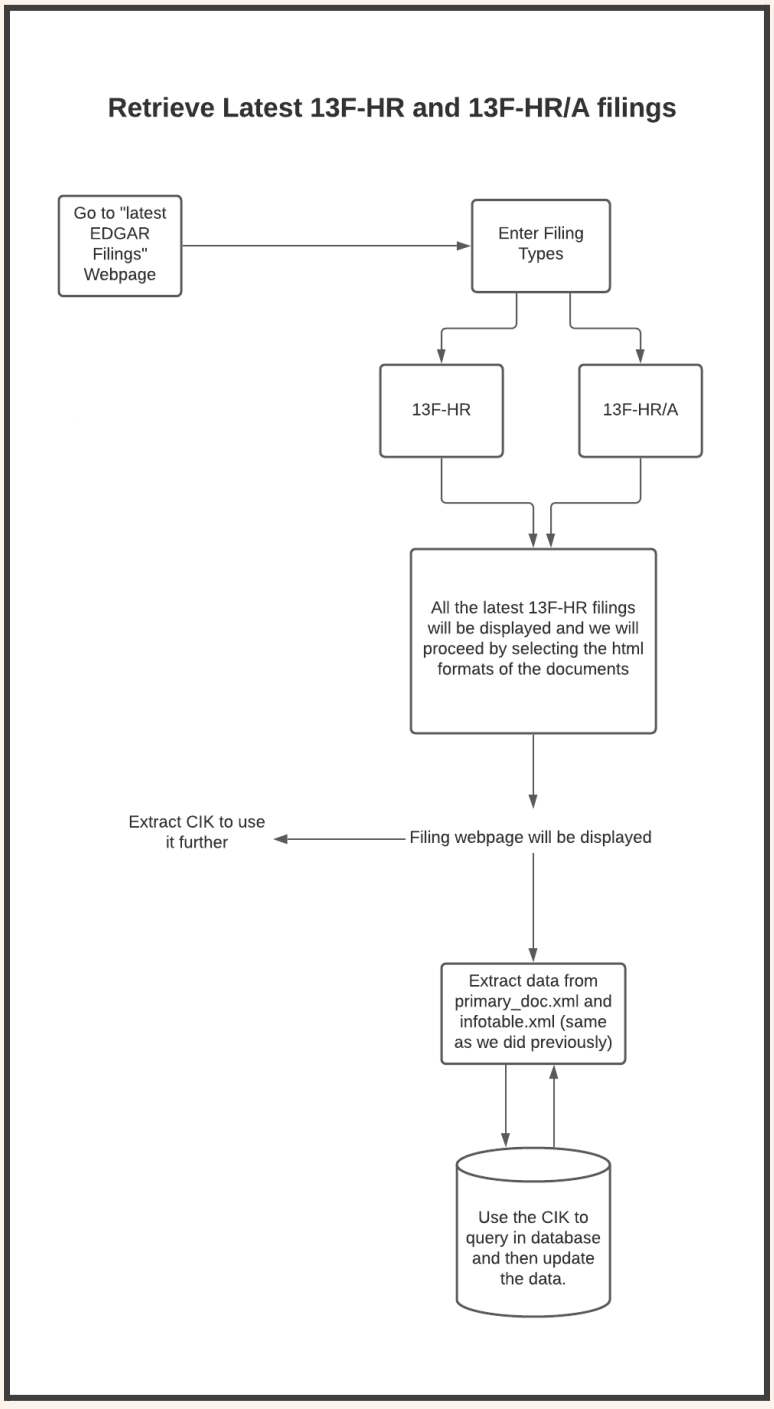
3.1.2 Get historic 13F data from EDGAR

A script to scrape data from EDGAR will run. The scraping script will retrieve data based on the CIKs and store it in the 13F database. The process for this is depicted below-



3.1.3 Retrieve latest 13F filings

Subsequent filings in the future would automatically be scraped by a script. The process is shown below-



3.1.4 Calculations : Filer View

This section aims to define the methodology for each of the calculated fields that will be used in the Filer view.

A. Previous Market Value

a. Calculated for each filer.

b. Market value from quarterlyholding table for previous quarter, adjusting for amendments.

B. Number of Previous Holdings

a. Calculated for each filer.

b. Number of Holdings from quarterlyholding table for previous quarter, adjusting for amendments.

C. Number of New Holdings

a. Calculated for each filer.

b. Count of holding identifiers marked as new for “Position Type” in Holdings View

D. Number of Increased Holdings

a. Calculated for each filer.

b. Count of all holdings marked as Increased for “ Position Type” in Holdings View of Current Quarter

E. Number of Decreased Holdings

a. Calculated for each filer.

b. Count of all holdings marked as Decreased for " Position Type" in Holdings View of Current Quarter

F. Number of Sold Out Holdings

a. Calculated for each filer.

b. Count of all holdings marked as Decreased for " Position Type" and quantity = 0 in Holdings View of Current Quarter

G. Top 10 Holdings Weight (%)

a. Calculated for each filer.

b. Sort Holdings View for current quarter in descending order by Weight or Market Value and sum the weights for first 10 holdings.

H. Average Holding Period

a. Calculated for each filer.

b. For the current quarter, average the held duration column.

I. Filer Description

a. Calculated for each filer.

b. Exact text to be decided but something like:

Description format-

Filer Name is a *Company Type* located out of *State*, *Country*. Their latest 13F filings show that they have at least *Market Value* AUM. They have long positions in *# of Holdings* with an average holding period of *Average Holding Period*. Their Biggest Holding is *Biggest Holding*, which makes up *Biggest Holding Weight* of their portfolio. Our database has at least *# of Filings* 13F filings of this filer.

Note: Underlined text refers to the fields that would be gathered from the database.

3.1.5 Visualizations

This section states the various visualizations that would be displayed in the application.

A. Individual Fund

a. Holdings as of any quarter end

b. Top Buys/Sells in current quarter

c. Holdings history for individual positions

d. Industry/Sector breakdown for each manager

e. Changes in industry/sector allocations over time

f. Top performing positions

g. Fund historical performance data

h. Fund Quality Statistics : Portfolio Volatility , Sharpe Ratio, Timing Accuracy etc.

i. Eventually something even more ambitious – Performance and Risk Attribution Models.

B. Aggregated Views

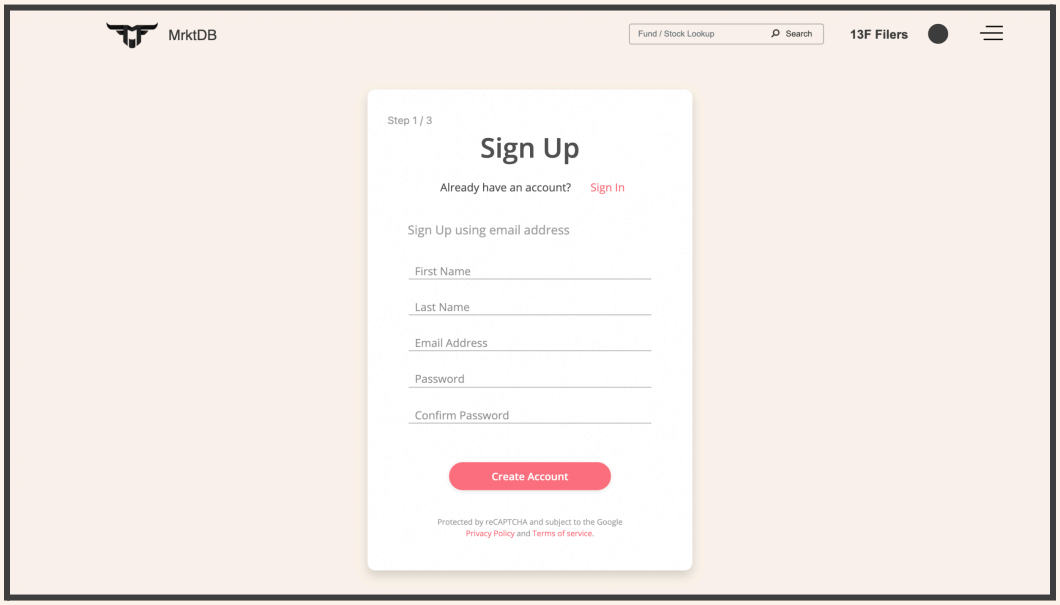
- **a.** Overall Hedge fund assets over time
- **b.** Compare hedge fund industry assets with other trends – Mutual fund assets, etf assets, etc.
- **c.** Hedge Fund Holdings Time series of individual stocks
- **d.** Individual Stocks – Number of funds holding the stock.
- **e.** Hedge Fund Holdings Time series of individual industries and sectors
- **f.** Hedge Fund performance by hedge fund types
- **g.** Hedge Funds Top Favorite Stocks – as of any quarter
- **h.** Hedge Funds New Favorite Stocks (most % increase in holdings of individual stocks)
- **i.** Stocks falling out of favor (QoQ largest decline in Hedge fund holdings)

3.2 Software Product Features

3.2.1 Sign Up Module

This function allows the user to sign up and undergo validation checks.

- ➔ User fills the registration forms
- ➔ Validation checks are performed for email ID syntax and password strength.
- ➔ Strong password recommendations are provided.
- ➔ Checks are made to see if the user already exists
  - ◆ if it exists, then the user is asked to sign in
  - ◆ else the user proceeds to the next step
- ➔ After sign up, an email is sent for account verification.
- ➔ The account is confirmed and user details are stored in the user database.
- ➔ The user can then sign in into the portal.



3.2.2 Sign In Module

This function allows the user to enter into the application. The user is required to provide an email and password. After authentication users will have access to most functionalities of the application. Availability of premium functionalities depend on the user's permission package(whether the user is a premium user or not).

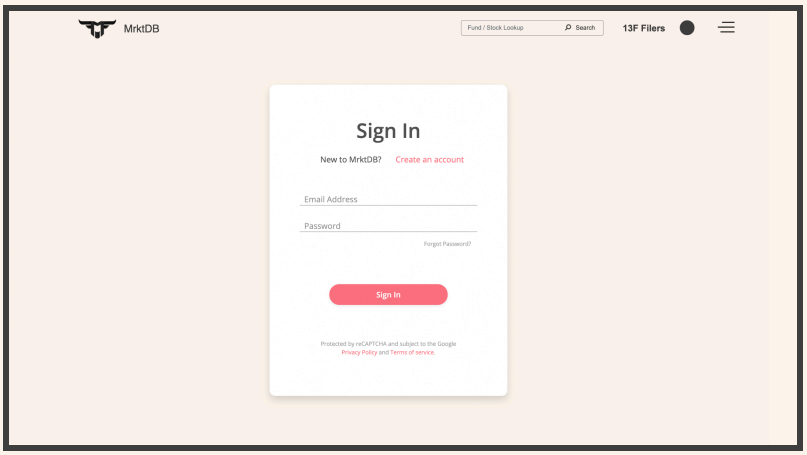
➔ User fills the Sign In form.

➔ Checks if the email already exists in the database.

- ◆ If the email exists then check the password
  - If password is correct the user is signed in
  - Else display an error message stating ‘Password do not match’
- ◆ Else display an error message stating ‘Credentials do not match’

➔ Forgot Password Feature

- ◆ User enters the registered email
- ◆ If the email exists, a reset password link is sent to the email Id
  - User enters the new password and new credentials are stored in the database
- ◆ Else an error stating ‘No user exists with the provided email’ is displayed



3.2.3 Navbar

➔ The product logo and name will be displayed in the navbar.

➔ Also, a search box will be provided for easy access to search for any Hedge Funds, FIlers, or stocks.

➔ User dashboard can also be accessed by clicking on the circular element which will display the user profile picture.

➔ A hamburger menu will contain links to all the pages on the website.



3.2.4 Hamburger menu

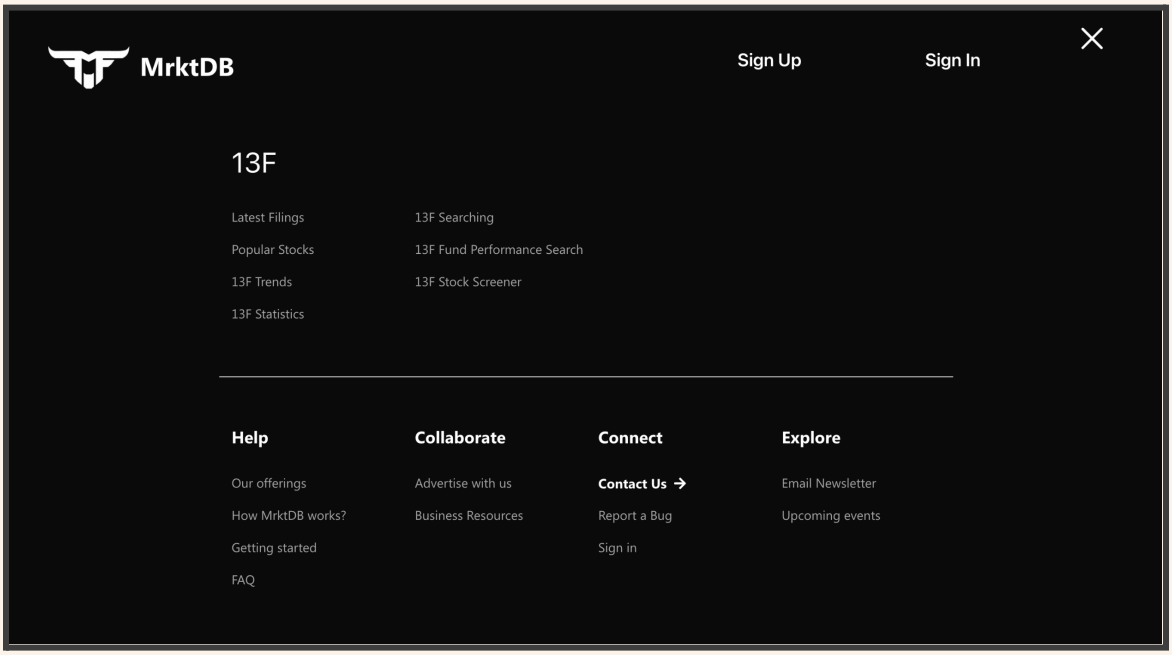
➔ The product logo and name will be displayed on the top left.

➔ Links to the Sign Up and Sign In pages will be placed on the top right.

➔ A cross button on the top right will be displayed to close the hamburger menu.

➔ Links to 13F filings, trends, and statistics will be displayed.

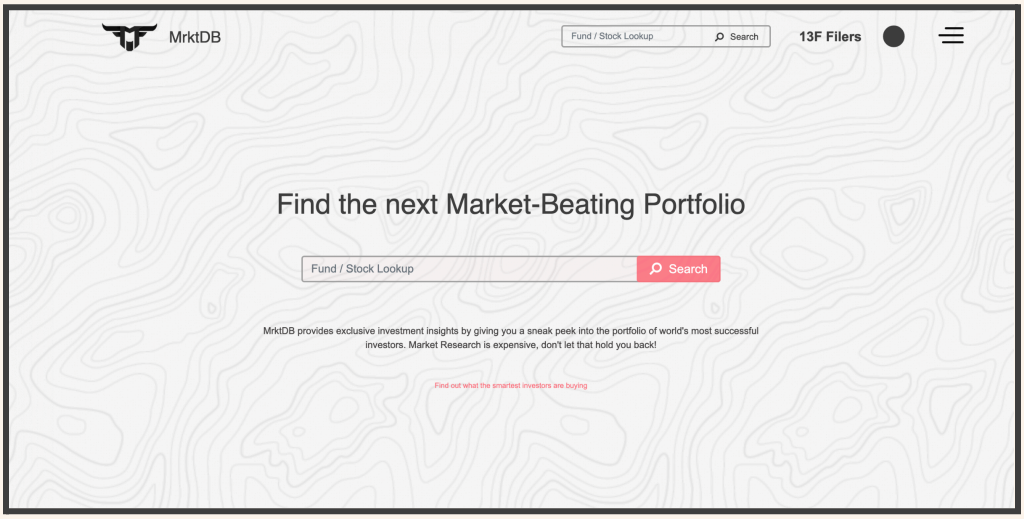
➔ Also, the user gets other options like FAQ, email newsletter, contact us, etc. as shown in the image below.



3.2.5 Search

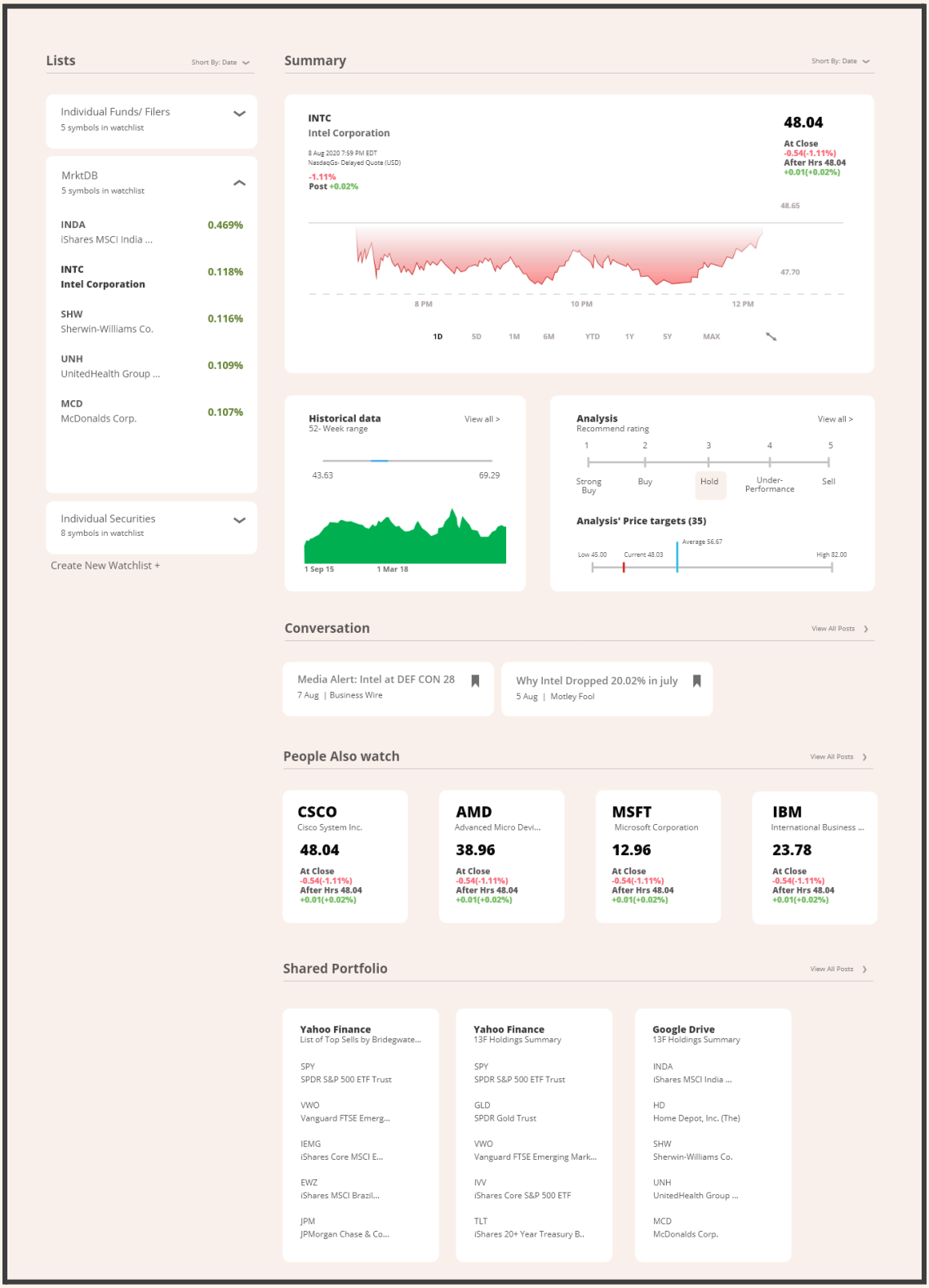
This function allows the user to search for any Hedge Funds, FIlers, or stocks and displays the page for the relevant search results.

- ➔ The main search box will be provided at the homepage where the user lands for the first time when the website is opened.
- ➔ A search box with the same functionality will also be provided in the navbar which was shown in section 3.1.3.



3.2.6 User Dashboard

- ➔ Add funds and individual stocks to watchlists.
- ➔ The user can view his/her watchlisted funds and stocks on the dashboard page.
- ➔ Charts related to the top fund/stock of the users' choice will also be shown here.
- ➔ Bookmarked items by the user will be displayed below the charts.
- ➔ Top 4 most popular stocks on MrktDB will be displayed below the bookmarked items.
- ➔ Provide customized insights about these watchlists.
    - ◆ Customized news about the items in watchlist
    - ◆ Aggregate performance of funds in the watchlist over time- A time series chart.
    - ◆ Sector/Industry Allocation of funds in the watch list – Time series bar graph.
    - ◆ Recommendations for buys or sells based on the quarterly activity of hedge funds.

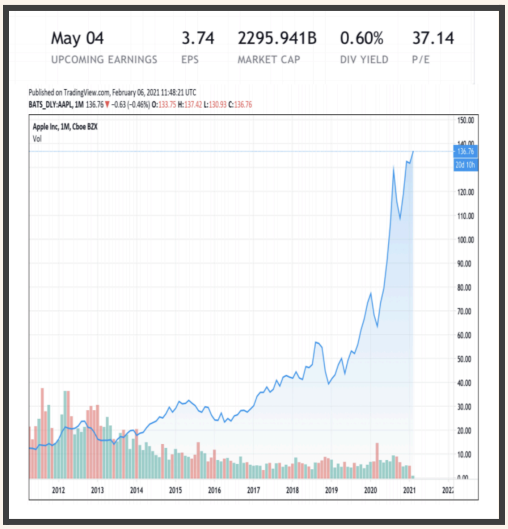

Lists
Short By: Date
Individual Funds/ Filers
5 symbols in watchlist
MrktDB
5 symbols in watchlist
INDA
iShares MSCI India ...
0.469%
INTC
Intel Corporation
0.118%
SHW
Sherwin-Williams Co.
0.116%
UNH
UnitedHealth Group ...
0.109%
MCD
McDonalds Corp.
0.107%
Individual Securities
8 symbols in watchlist
Create New Watchlist +
Summary
Short By: Date
INTC
Intel Corporation
8 Aug 2020 7:59 PM EDT
NasdaqGs- Delayed Quote (USD)
-1.11%
Post +0.02%
48.04
At Close
-0.54(-1.11%)
After Hrs 48.04
+0.01(+0.02%)
48.65
47.70
8 PM
10 PM
12 PM
1D
5D
1M
6M
YTD
1Y
5Y
MAX
Historical data
52- Week range
View all >
43.63
69.29
1 Sep 15
1 Mar 18
Analysis
Recommend rating
View all >
1
2
3
4
5
Strong Buy
Buy
Hold
Under-Performance
Sell
Analysis' Price targets (35)
Low 45.00
Current 48.03
Average 56.67
High 82.00
Conversation
View All Posts
Media Alert: Intel at DEF CON 28
7 Aug | Business Wire
Why Intel Dropped 20.02% in july
5 Aug | Motley Fool
People Also watch
View All Posts
CSCO
Cisco System Inc.
48.04
At Close
-0.54(-1.11%)
After Hrs 48.04
+0.01(+0.02%)
AMD
Advanced Micro Devi...
38.96
At Close
-0.54(-1.11%)
After Hrs 48.04
+0.01(+0.02%)
MSFT
Microsoft Corporation
12.96
At Close
-0.54(-1.11%)
After Hrs 48.04
+0.01(+0.02%)
IBM
International Business ...
23.78
At Close
-0.54(-1.11%)
After Hrs 48.04
+0.01(+0.02%)
Shared Portfolio
View All Posts
Yahoo Finance
List of Top Sells by Bridegwate...
SPY
SPDR S&P 500 ETF Trust
VWO
Vanguard FTSE Emerg...
IEMG
iShares Core MSCI E...
EWZ
iShares MSCI Brazil...
JPM
JPMorgan Chase & Co...
Yahoo Finance
13F Holdings Summary
SPY
SPDR S&P 500 ETF Trust
GLD
SPDR Gold Trust
VWO
Vanguard FTSE Emerging Mark...
IVV
iShares Core S&P 500 ETF
TLT
iShares 20+ Year Treasury B..
Google Drive
13F Holdings Summary
INDA
iShares MSCI India ...
HD
Home Depot, Inc. (The)
SHW
Sherwin-Williams Co.
UNH
UnitedHealth Group ...
MCD
McDonalds Corp.

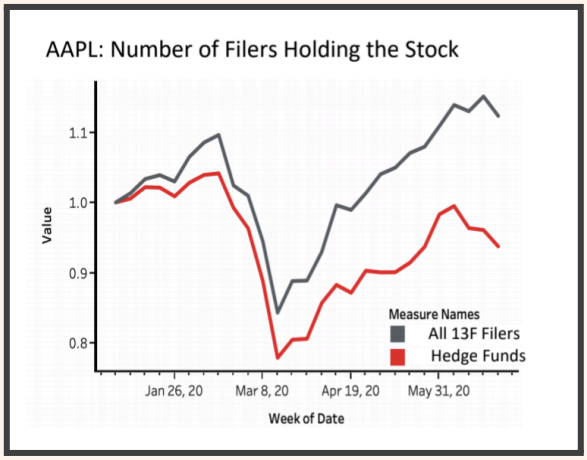
3.2.7 Display Charts

This function allows the user to view charts based on the data retrieved from the database.

➔ Charts showing the trends like number of filers holding the stock will be plotted against the date. Other types of charts will also be shown on the individual stock dashboard.

➔ Also, some charts will be embedded from TradingView.





3.2.8 Display Tables

This function allows the user to view tables based on the data retrieved from the database.

➔ Tables will be displayed on the individual stock and individual fund dashboard.

➔ Other tables will be displayed in the 13F statistics section.

➔ Some tables will be embedded from TradingView.

Symbol	Security Name	Security Type	Market Value change (Trading)	Market Value change% (Trading)	★
LBTYA	LIBERTY GLOBAL PLC	STOCK	-7,941,314	-2.43%	☆
AMZN	AMAZON COM INC	STOCK	-7,798,884	-0.74%	★
LSXMK	LIBERTY MEDIA CORP DELAWARE	STOCK	-7,588,970	-0.77%	☆
AAL	AMERICAN AIRLS GROUP INC	STOCK	-7,204,299	-1.39%	☆
GM	GENTAL MTRS CO	STOCK	-6,628,836	-0.43%	☆

Download data Add to watchlist Show more

Stock		Sector	Shares Held or Principal Amt	Market Value	% of Portfolio	Previous % of Portfolio	Rank	Change in Shares	% Change	% Ownership
AAPL		INFORMATION TECHNOLOGY	245,155,566	$62,340,609,000	36.49	29.74	1	No Change	0%	5.6029%
BAC		FINANCE	925,008,600	$19,637,932,000	11.50	13.46	2	No Change	0%	10.6623%
KO		CONSUMER STAPLES	400,000,000	$17,700,001,000	10.36	9.15	3	No Change	0%	9.3134%
AXP		FINANCE	151,610,700	$12,979,391,000	7.60	7.80	4	No Change	0%	18.8343%
WFC		FINANCE	323,212,918	$9,276,210,000	5.43	7.18	5	No Change	0%	7.8902%
KHC		CONSUMER STAPLES	325,634,818	$8,056,205,000	4.72	4.32	6	No Change	0%	26.6530%
MCO		FINANCE	24,669,778	$5,217,658,000	3.05	2.42	7	No Change	0%	13.1778%
JPM		FINANCE	57,714,433	$5,196,030,000	3.04	3.43	8	↓1,800,499	-3%	1.8944%
USB		FINANCE	150,972,301	$5,031,906,792	2.95	2.60	9	↑18,512,683	13%	9.9242%
DVA		HEALTH CARE	38,095,570	$2,897,549,000	1.70	1.20	10	↓470,000	-1%	31.2757%

3.2.9 13F Trends

This function helps in showing various historic trends by analysing the data. The charts will show the amount of shares owned by a company over time. It would help users in better decisions while investing in the market.



3.2.10 Individual Fund Dashboard

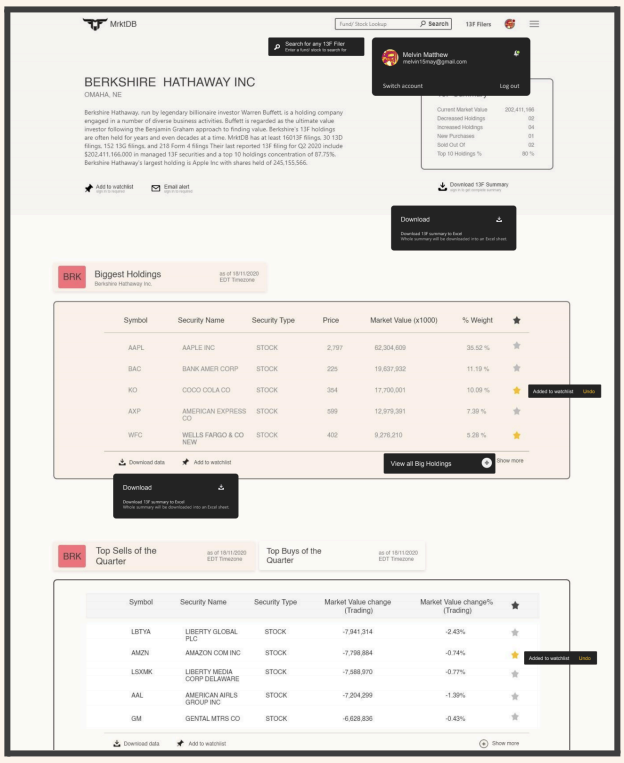
When the user will search a hedge fund's name, it will take the user to the specific fund's dashboard. This dashboard format will stay consistent across 8000+ hedge funds that will be tracked. Historic data about the holdings by the hedge funds will be displayed.

The dashboard will show the following:

- ➔ Hedge Fund name, description, and associated details like strategy, manager, location, etc.
- ➔ A quick summary table with metrics like Market value, number of positions, number of new buys/sells, etc.
- ➔ A news feed related to the hedge fund and its holdings.
- ➔ There will be basic summary tables like
    - ◆ Top Holdings
    - ◆ Top Buys
    - ◆ Top Sells

➔ There will be basic visualizations like

- ◆ Performance over time compared to S&P 500
- ◆ Industry/sector Allocation
- ◆ Small/Mid/Micro-Cap Allocation





3.2.11 Single Stock Dashboard

When the user will search a stock's ticker/name, it will be redirected to the specific stock's dashboard. This dashboard format will stay consistent across 10,000+ stocks that will be tracked.

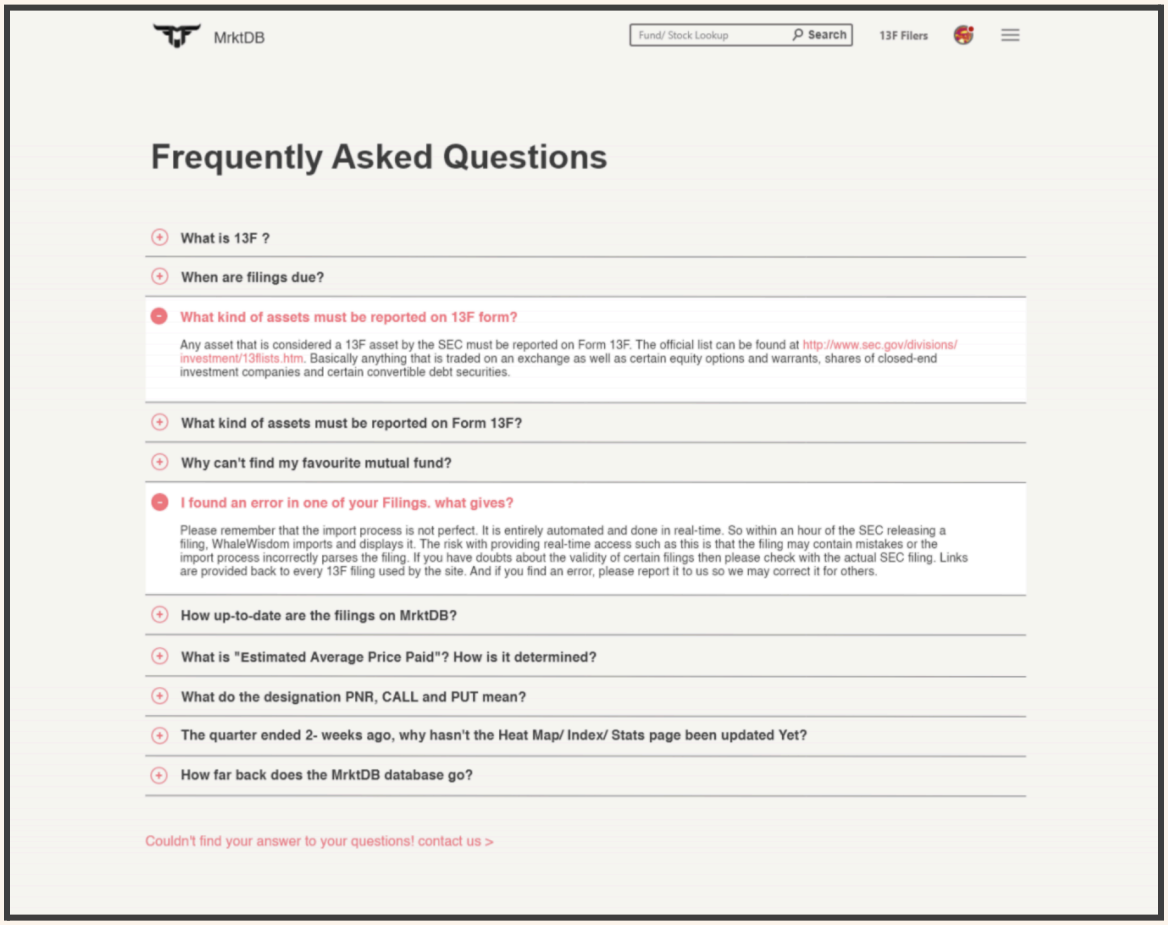
The dashboard will show the following:

- ➔ Stock's name, description, and associated details like industry, location, etc.
- ➔ A quick summary table with metrics like Market cap, shares outstanding, average volume, bid-ask spread, P/E ratio, etc.
- ➔ Market Data visualization – Most likely embedded charts from other vendors like TradingView that show the time series of price, volume, etc.
- ➔ Aggregated 13F data table showing things like aggregated number of shares held across hedge funds, number of new buys/sells, stock's popularity rating, etc
- ➔ Aggregated 13F Visualizations like
    - ◆ Stock's holdings over time across all hedge funds vs. returns over time. (line chart)
    - ◆ Stock's new buys/sells during each historical quarter (bar chart)
- ➔ A table showing funds with the biggest holdings of this stock.



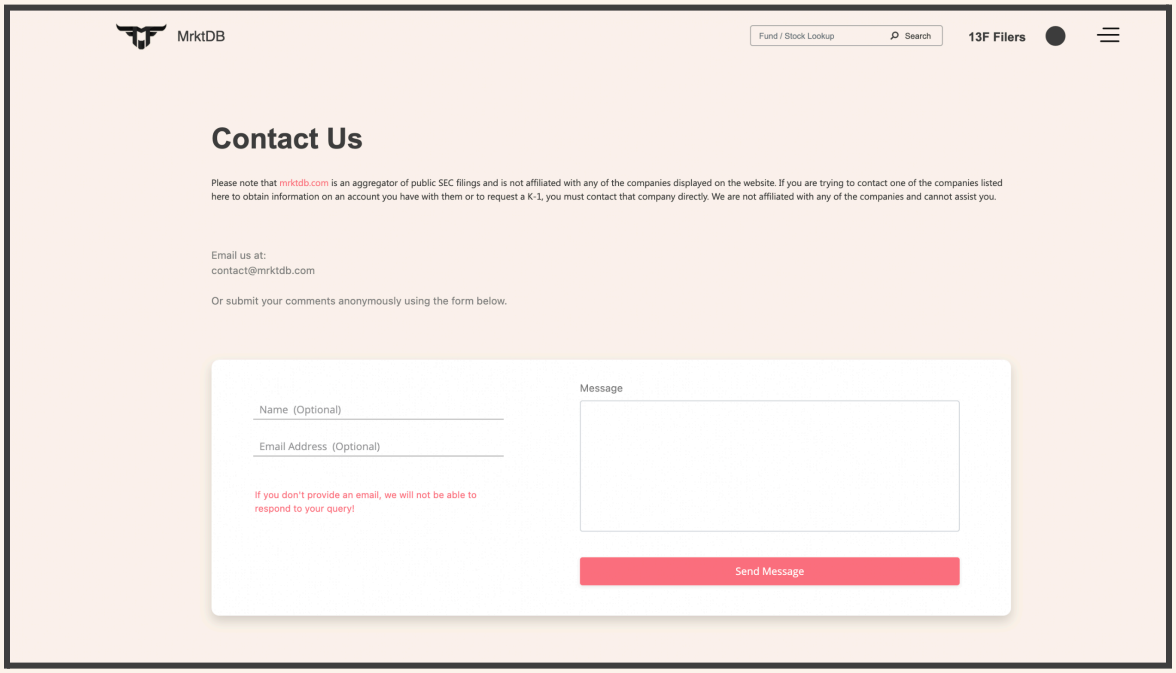
3.2.12 FAQ

The FAQ functionality in the application is meant to resolve basic doubts of the users about the website. The admin will put up answers to general questions about the website in this section. The users can also send their own questions and if the admin feels that it is really sensible, then it can be added to the FAQ page.



3.2.13 Contact / Help

Contact/Help function allows users to send an email to the admin in case of any queries/suggestions for the website.

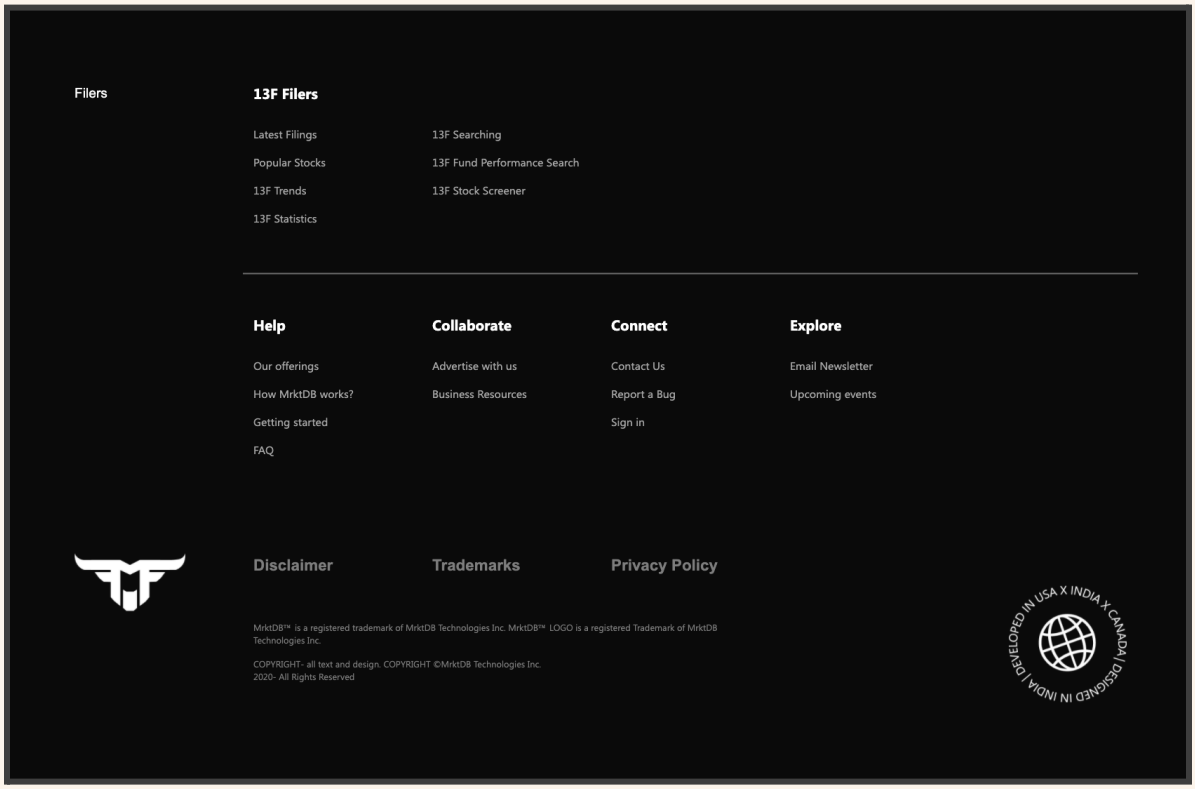


3.2.14 Create / Print Reports

Each 13F filer data can be exported/downloaded by any registered user in csv format. This would enable users to work with the data on their own if they want to.

3.2.15 Footer

The footer will look almost similar like the hamburger menu. In addition, a few elements like logos and links to the privacy policy, disclaimer, and trademarks will also be shown.



3.2.16 Logout

This function can be performed by registered users. It terminates the user session. The system can also perform this function automatically if the session is left unused for half an hour.

3.3 Performance requirements

➔ System should respond and generate output within 3 seconds when the user interacts with the system, else a message defining the error should be shown.

➔ An efficient web scraper that would run every night to look for any new 13F filings on EDGAR.

➔ Charts should be generated within 2 seconds.

3.4 Design constraints

➔ A max-width of 1500px would be applied to the website contents. So, users on extra large screens will have to zoom in to view the website properly.

3.5 Software System Attributes

3.5.1 Reliability

The application should deliver appropriate and correct market data to the users. Any false information displayed would result in dissatisfaction from the users. It should also be mentioned that the market data displayed on the website is not real-time data.

3.5.2 Security

The server on which the application resides will have its own security to prevent unauthorized *write/delete* access. There is no restriction on read access.

The system on which the user uses the application will have its own security. There is no special protection built into this system other than that the information of the users shall be encrypted before transferred to the database in order to maintain the security of the crucial information about users.

3.5.3 Maintainability

Maintainability of the application would be a little hard sometimes as there are several dependencies and the tools being used are continuously being updated by developers. So, frequent checks would have to be made to ensure proper and desired functioning of the application.

3.6 Database Requirements

3.6.1 User Database

User database will consist of some required fields which include-

- First Name : A character field with max length = 100
- Last Name : A character field with max length = 100
- Email : An email field with max length = 100
- Password : A character field with max length = 200. The password would be stored as a hash value for security reasons.

The user can also fill some non-required fields if he/she wants to provide his/her information willingly.

- Occupation : A character field with max length = 100
- Company : A character field with max length = 100
- City : A character field with max length = 50
- State : A character field with max length = 50
- Phone number : A numeric field with length = 10

An example of the user database schema is shown below-

User Database								
First Name	Last Name	Email	Password	Occupation	Company	City	State	Phone
Ashutosh	Bhardwaj	abhardwaj@mrktdb.com	25jn325jbi235235253k5	Risk Analyst	State Street	Austin	Texas	-
Melvin	Mathew	mmathew@mrktdb.com	gs3352532623nk26523	Software Engineer	HubSpot	-	-	011-254-4554

3.6.2 13F Database

The 13F database will consist of data scraped from edgar. It will include the following tables-

Company

- companyId : Primary key for Company table
- CIK : A Text-field
- name : A Text-Field with max length = 80
- address : A Text-Field with max length = 120
- companyType : A character field with max length = 2
- createdAt : DateTime field with auto values
- updatedAt : DateTime field with auto values
- deletedAt : DateTime field with auto values

Company							
companyId	CIK	name	address	type	createdAt	updatedAt	deletedAt
10	1004244	NEW ENGLAND ASSET MANAGEMENT INC		IA	1594503053976	1594503053976	
20	352532	GENERAL RE CORP		HC	1594503702512	1594503702512	
30	1067983	BERKSHIRE HATHAWAY INC	3555 Farnam Street Omaha, NE 68131	HF	1594503906559	1594506762311	
40	5326266	APPLE INC		HC	1594505095430	1594505095430	
50	315090	Buffett Warren E	3555 Farnam St STE 1440 Omaha, NE 68131	IA	1594506897176	1594507584076	

Filer

- filerId : Primary key for Filer table
- companyId : Foreign key to Company table
- fileNumber : A character field with max length = 10
- filerType : A character field with max length = 3
- createdAt : DateTime field with auto values
- updatedAt : DateTime field with auto values
- deletedAt : DateTime field with auto values

Filer						
filerId	companyId	file number	type	createdAt	updatedAt	deletedAt
1	10	028-05194	13F	1594503053976	1594503053976	
2	20	028-04922	13F	1594503702512	1594503702512	
3	30	028-04545	13F	1594503906559	1594506974306	
4	50	028-00554	13F	1594506974306	1594506974306	

Quarterly Holding

- filerId : Foreign key to Filer table
- quarter : Text field indicating the quarter
- filingType : A character field with max length = 3
- filedOn : DateTime field
- acceptedAt : DateTime field
- totalValue : A Float field
- totalEntry : A Float field
- createdAt : DateTime field with auto values
- updatedAt : DateTime field with auto values
- deletedAt : DateTime field with auto values

Quarterly Holding

filerId	quarter	filing type	filedOn	acceptedAt	totalValue	totalEntry	createdAt	updatedAt	deletedAt
1	12020	HR	1589515200000	1589574876000	353520	53532	1594503053976	1594503053976	
3	12020	CR	1589515200000	1589573206000	254546	53352	1594507166324	1594507166324	
4	12020	NT	1589515200000	1589573344000	5352523	253532	1594507726378	1594507726378	

Quarterly Other Manager

- quarterlyHoldingId : Foreign key to Quarterly Holding table
- otherManagerId : Primary key
- parentFilerId : Foreign key to Filer table
- childFilerId : Foreign key to Filer table
- number : An Integer field
- createdAt : DateTime field with auto values
- updatedAt : DateTime field with auto values
- deletedAt : DateTime field with auto values

Quarterly Other Manager							
quarterlyHoldingId	otherManagerId	parentFilerId	childFilerId	number	createdAt	updatedAt	deletedAt
4	1	1	2	1	1594503702512	1594503702512	
5	2	1	3	2	1594503906559	1594503906559	
2	3	3	4	4	1594507251768	1594507251768	

Security

- companyId : Foreign key to Company table
- securityId : Primary key
- cikCusipMappingId : A Text field
- cusip : A Text field
- securityName : A Text field
- securityType : A Text field
- ticker : Character field with max length = 5
- titleOfClass : A Text field
- createdAt : DateTime field with auto values
- updatedAt : DateTime field with auto values
- deletedAt : DateTime field with auto values

Security										
securityId	cusip	security name	companyId	securityType	ticker	titleOfClass	cikCusipMapping Id	createdAt	updatedAt	deletedAt
100	37833100	APPLE INC	40	STOCK	AAPL	COM	953222389	1594505193824	1594505193824	

Quarterly Security Holding

- uin : A Text field
- quarterlySecurityHoldingId : Primary key
- securityId : Foreign key to Security table
- quarterlyHoldingId : Foreign key to Quarterly Holding table
- marketvalue : A Float field
- quantity : A Float field
- holdingType : A Text field
- investmentDiscretion : Character field with max length = 4
- sole : Integer Field
- shared : Integer Field
- none : Integer Field
- createdAt : DateTime field with auto values
- updatedAt : DateTime field with auto values
- deletedAt : DateTime field with auto values

Quarterly Security Holding

uin	quartorlySecurityHoldingId	securityId	quarterId	marketvalue	quantity	type (SH, PRN, PUT , CALL)	Investment Discretion (SOLE, DFND, OTHER)	Sole	Shared	None	createdAt	updatedAt	deletedAt
43kj34	1000	100	1	1452250000	5711000	SH	DFND	0	5711000	0	1594505836231	1594505836231	
364634hn	1001	100	1	3578000	14071	SH	SOLE	14071	0	0	1594505900124	1594505900124	
64jk3b4	1002	100	1	43992000	173000	SH	DFND	173000	0	0	1594507372483	1594507372483	
646jb3364	1003	100	1	244118000	960000	SH	DFND	960000	0	0	1594507372483	1594507372483	

Quarterly Other Manager Distribution

- quarterlyOtherManagerDistributionId : Primary key
- quarterlySecurityHoldingId : Foreign key to Quarterly Security Holding table
- quarterlyOtherManagerId : Foreign key to Quarterly Other Manager table
- createdAt : DateTime field with auto values
- updatedAt : DateTime field with auto values
- deletedAt : DateTime field with auto values

Quarterly Other Manager Distribution					
quarterlyOtherManagerDistributionId	quartorlySecurityHoldingId	quarterlyOtherManagerId	createdAt	updatedAt	deletedAt
1111	1000	1	1594505836231	1594505836231	
1112	1000	2	1594505900124	1594505900124	
1113	1002	3	1594507480045	1594507480045	
1114	1003	3	1594507480045	1594507480045	

Fails To Deliver

- quantity : Integer Field
- description : A Text field
- cusip : A Text field
- ticker : Character field with max length = 7
- settlementDate : DateTime field
- createdAt : DateTime field with auto values

Fails To Deliver					
settlementDate	cusip	ticker	createdAt	quantity	description
1594505836231	1000	1	1594505836231	57423	A...
1594505833523	4523	2	1594505900124	46364	T...

3.7 Data Dictionary

Company					
Attribute	**Data Type**	**Size**	**Constraint**	**Description**	**Example**
Name	VARCHAR	80	NOT NULL	Registered Name of the Company	Berkshire Hathaway
CIK	VARCHAR	10	NOT NULL	Central Index Key (Field used by EDGAR to uniquely identify companies)	0001067983
Company_Id	INT	10	Primary Key	Unique Id for every Company created	1
Address	VARCHAR	150	NOT NULL	Registered Address of the Company	3555 Farnam Street Omaha, NE 68131
Company_Type	CHAR	2	NOT NULL	Type of Company (Hedge Fund, Investment Advisor, Holdings Company)	HF

QuarterlyHolding					
Attribute	**Data Type**	**Size**	**Constraint**	**Description**	**Example**
Filer_Id	INT	10	Foreign Key	Reference to the primary key of Filer Database	10
Quarter	INT	5	NOT NULL	Quarter (Example, Quarter 4 of 2020 - 20204)	20204
Filing_Type	CHAR	3	NOT NULL	Type Of Filing (CR, CRA, NT, NTA, HR, HRA)	HR
Filed_On	DATE	-	NOT NULL	Date of Filing	1518366756000
Accepted_At	DATE	-	NOT NULL	Date of Acceptance of Filing	1518366776000
Total_Value	INT	13	NOT NULL	Total value of holdings entries reported	353520
Total_Entry	INT	5	NOT NULL	Total number of entries reported	53532
Quarterly_Holding_Id	INT	10	Primary Key	Unique Id for every Quarterly Holding created	4

Filer					
Attribute	**Data Type**	**Size**	**Constraint**	**Description**	**Example**
Company_Id	INT	10	Foreign Key	Reference to the primary key of Company table	1
File_Number	VARCHAR	10	NOT NULL	File Number of the filing	028-05194
Filer_Id	INT	10	Primary Key	Unique Id for every filer created	10
File_Type	CHAR	3	NOT NULL	Type of Filing (13F/13G/13D)	13F

QuarterlyOtherManager					
Attribute	**Data Type**	**Size**	**Constraint**	**Description**	**Example**
Parent_Filer_Id	INT	10	Foreign Key	Reference to the primary key of Filer Database (Parent)	2
Child_Filer_Id	INT	10	Foreign Key	Reference to the primary key of Filer Database (Child)	4
Quarterly_Holding_Id	INT	10	Foreign Key	Reference to the primary key of Quarterly Holding Database	3
Number	INT	10	NOT NULL	Serial Number Of The Manager in the primary Doc Managers Table	1
Other_Manager_Id	INT	10	Primary Key	Unique Id for every Quarterly Other Manager created	11

Security					
Attribute	**Data Type**	**Size**	**Constraint**	**Description**	**Example**
Company_Id	INT	10	Foreign Key	Reference to the primary key of Security Database	40
CUSIP	INT	10	NOT NULL	Committee on Uniform Securities Identification Procedures Identification Number for the security	37833100
Security_Name	INT	80	NOT NULL	Name Of the Security	Apple Inc
Security_Type	CHAR	20	NOT NULL	Type Of Security	Stock
Ticker	CHAR	5	NOT NULL	Ticker Symbol of the Security	AAPL
Title_Of_Class	INT	20	NOT NULL	Title Of Class	COM
Security_Id	INT	10	Primary Key	Unique Id for every Security created	100

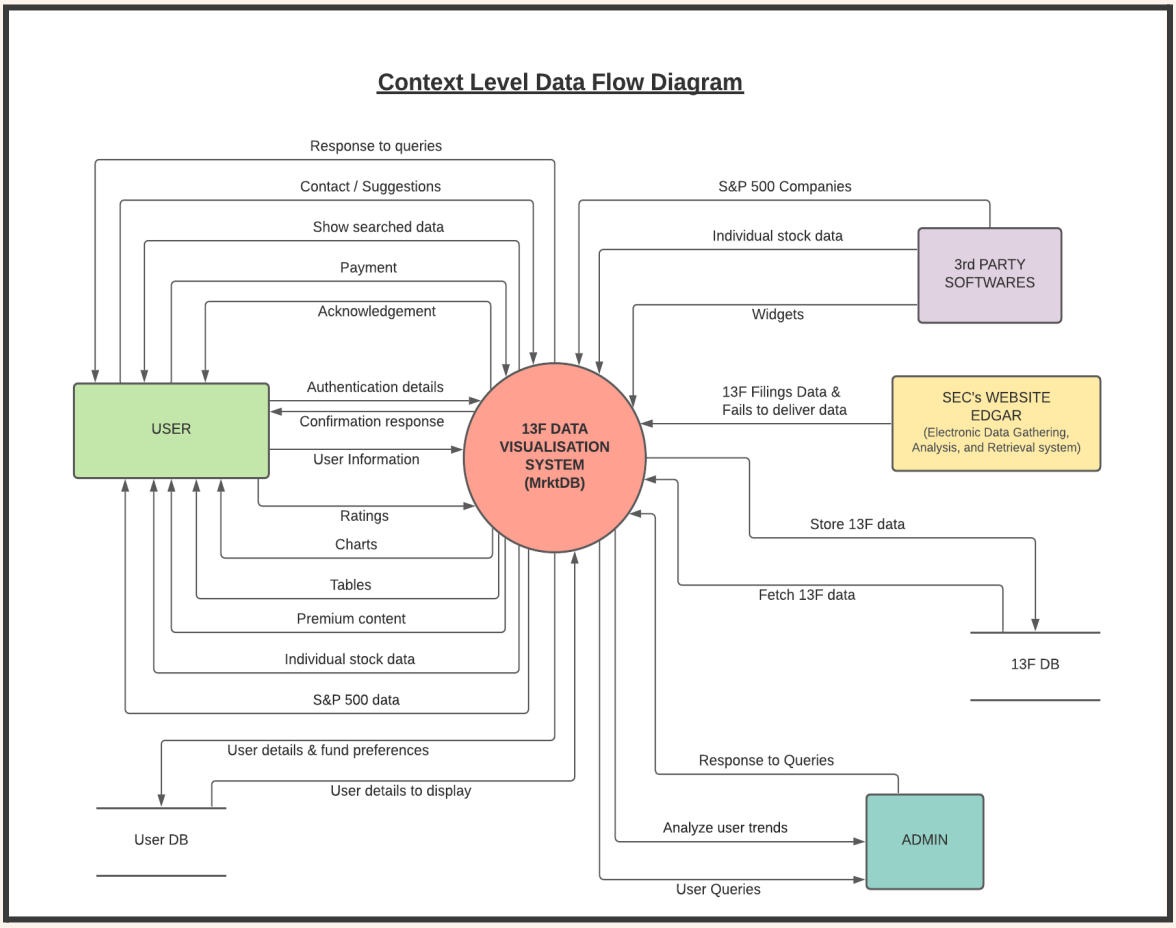
QuarterlySecurityHolding					
Attribute	**Data Type**	**Size**	**Constraint**	**Description**	**Example**
securityId	INT	10	Foreign Key	Reference to the primary key of Security Database	100
quarterlyHoldingId	INT	10	Foreign Key	Reference to the primary key of Quarterly Holding Database	1000
marketvalue	INT	13	NOT NULL	Market Value Of the Security Reported By the filer	3568000
quantity	INT	10	NOT NULL	Quantity Of the Security Reported By the filer	14071
holdingType	CHAR	4	NOT NULL	Type Of Holding (SH, PRN, PUT, CALL)	SH
investmentDiscretion	CHAR	4	NOT NULL	Investment Discretion for the Holding (SOLE, DFND, OTHR)	DFND
sole	INT	10	NOT NULL	Number Of Sole Holdings	14071
shared	INT	10	NOT NULL	Number Of Shared Holdings	0
none	INT	10	NOT NULL	Number Of None Holdings	0
quarterlySecurityHoldingId	INT	10	Primary Key	Unique Id for every Quarterly Security Holding created	563

QuarterlyOtherManagerDistribution					
Attribute	**Data Type**	**Size**	**Constraint**	**Description**	**Example**
Quarterly_Other_Manager_Id	INT	10	Foreign Key	Reference to the primary key of Quarterly Other Manager Database	100
Quarterly_Security_Holding_Id	INT	10	Foreign Key	Reference to the primary key of Quarterly Security Holding Database	2201
Quarterly_Other_Manager_Distribution_Id	INT	10	Primary Key	Unique Id for every Quarterly Other Manager Distribution created	432

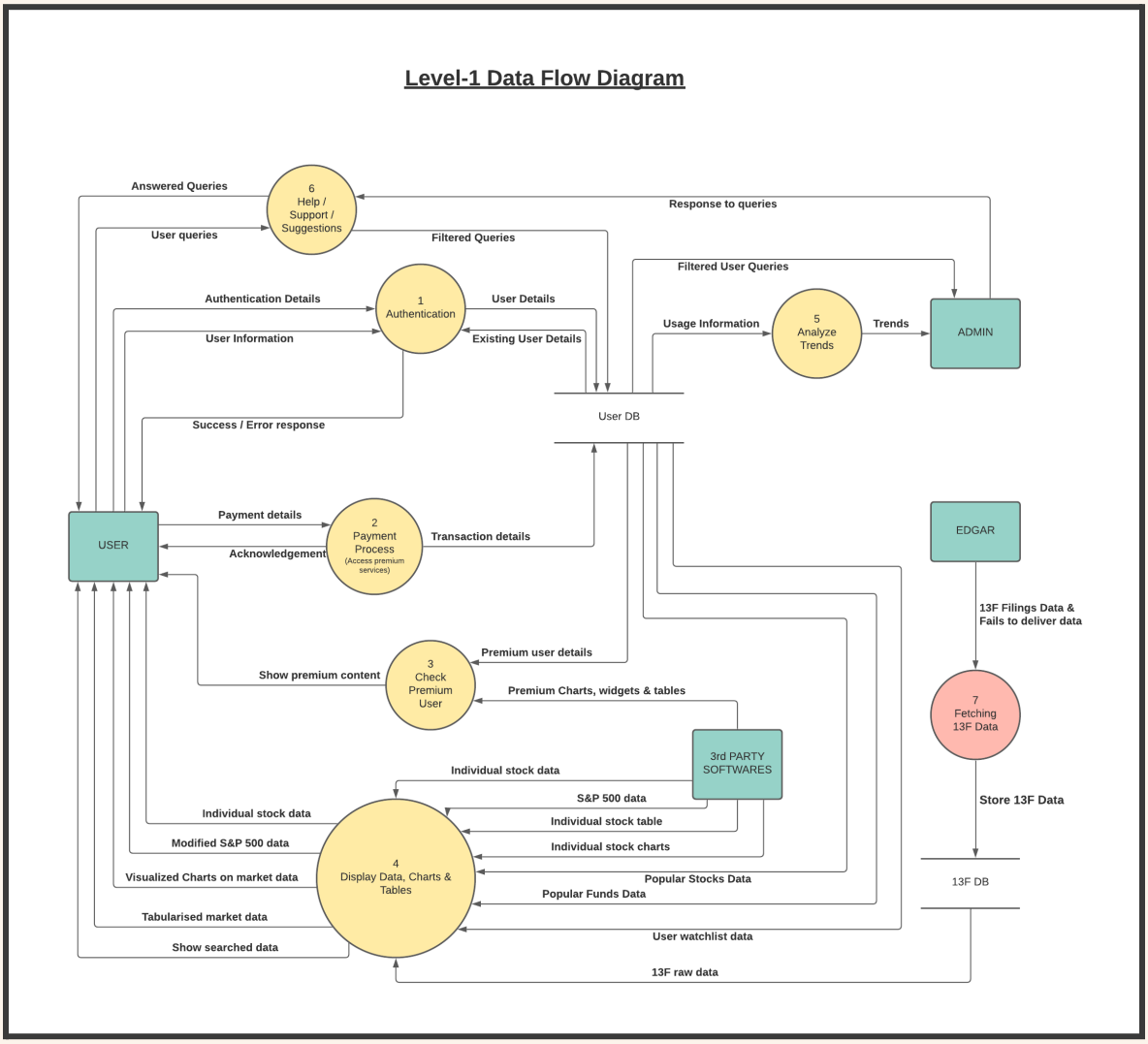
4. Supporting Information

4.1 Data Flow Diagrams

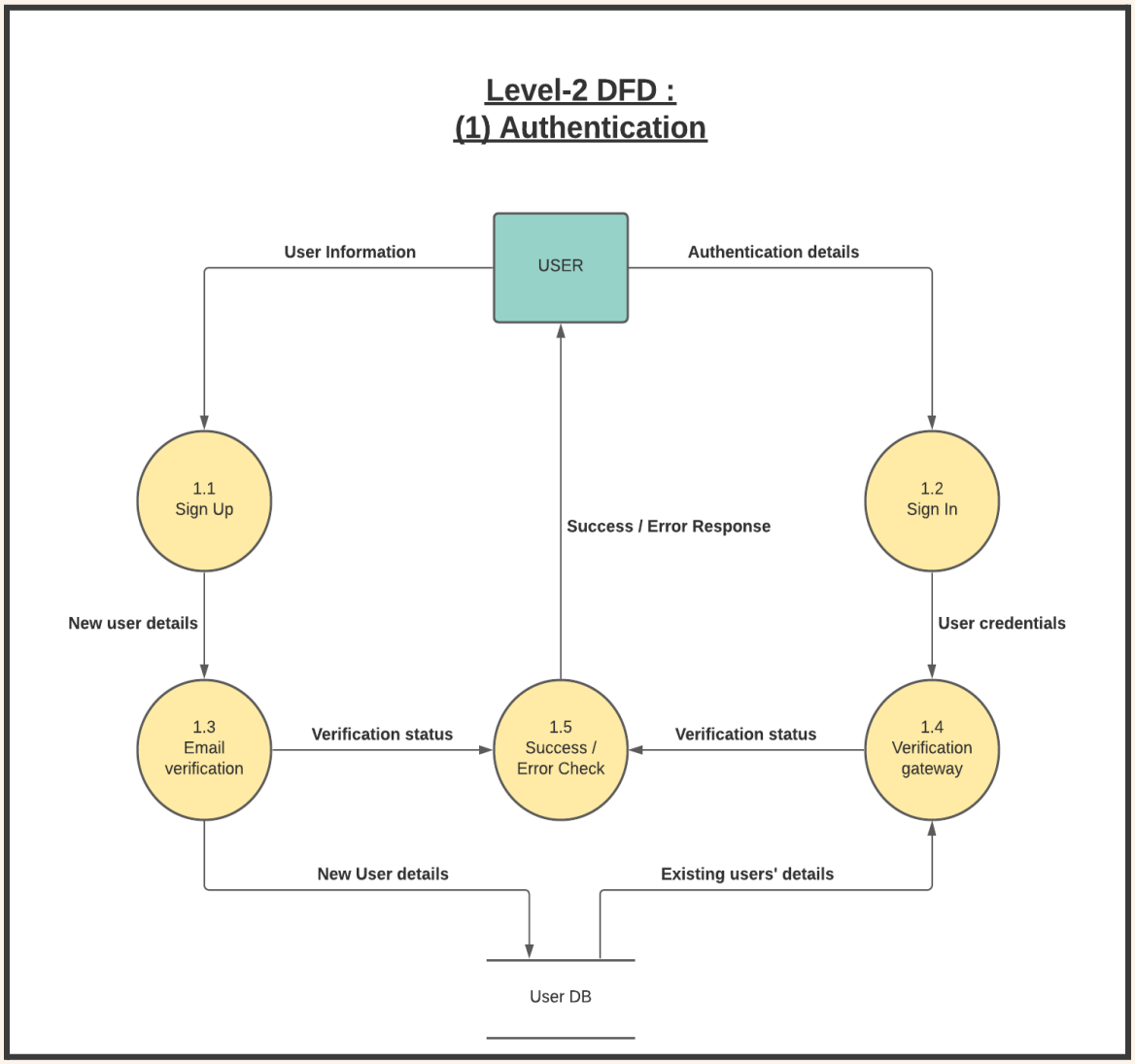
4.1.1 Context level DFD



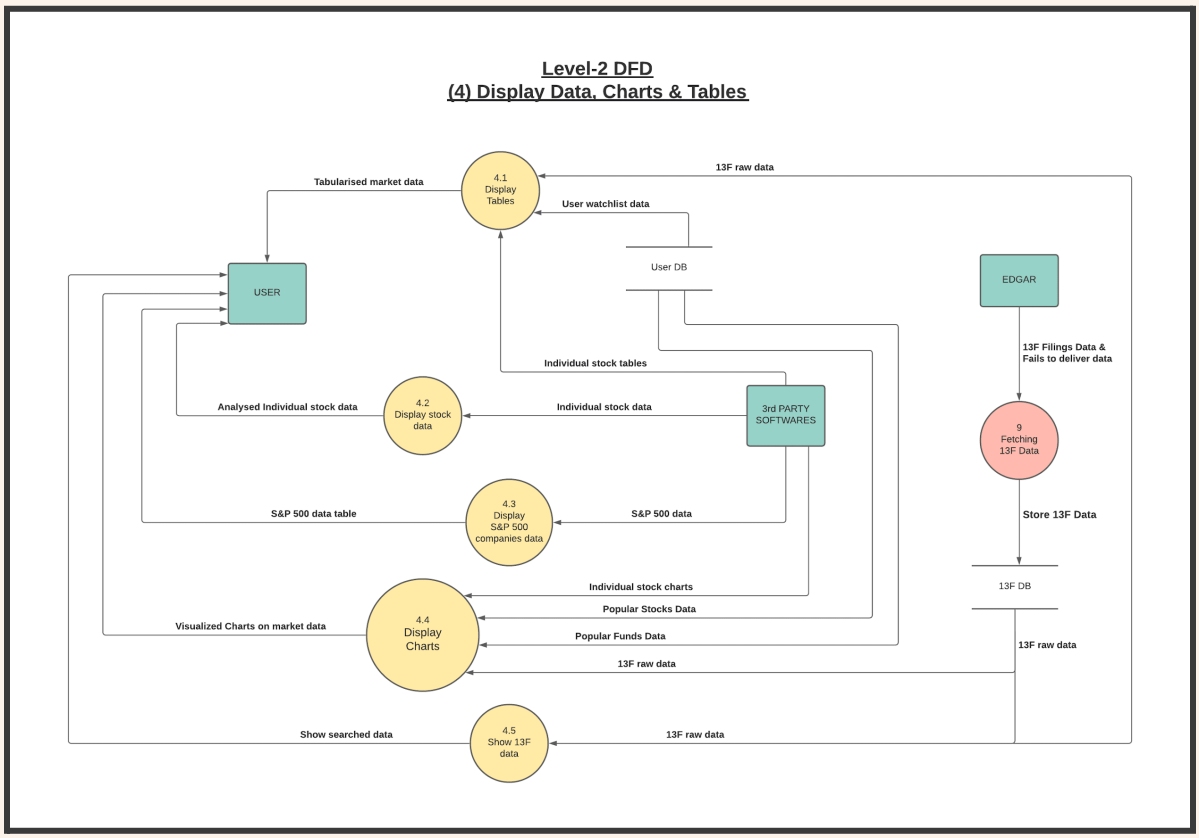
4.1.2 Level-1 DFD



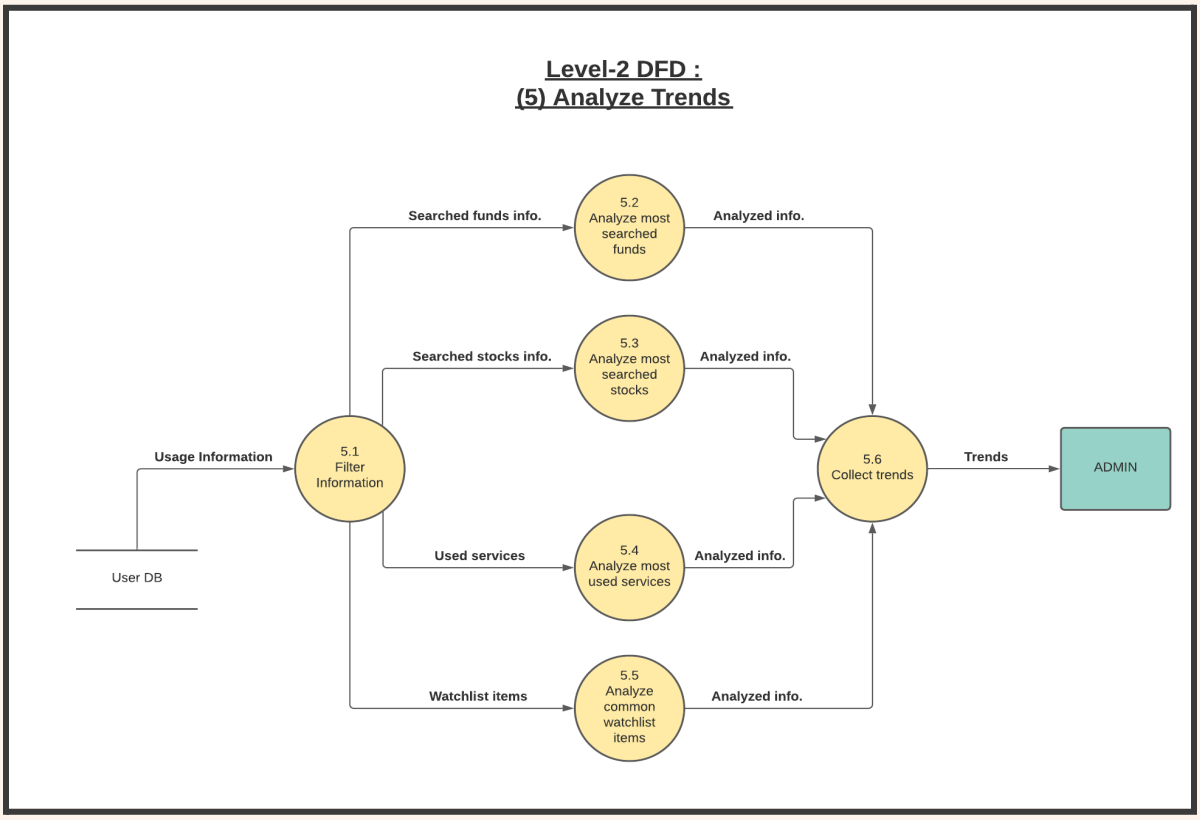
4.1.3 Level-2 DFD (1. Authentication)



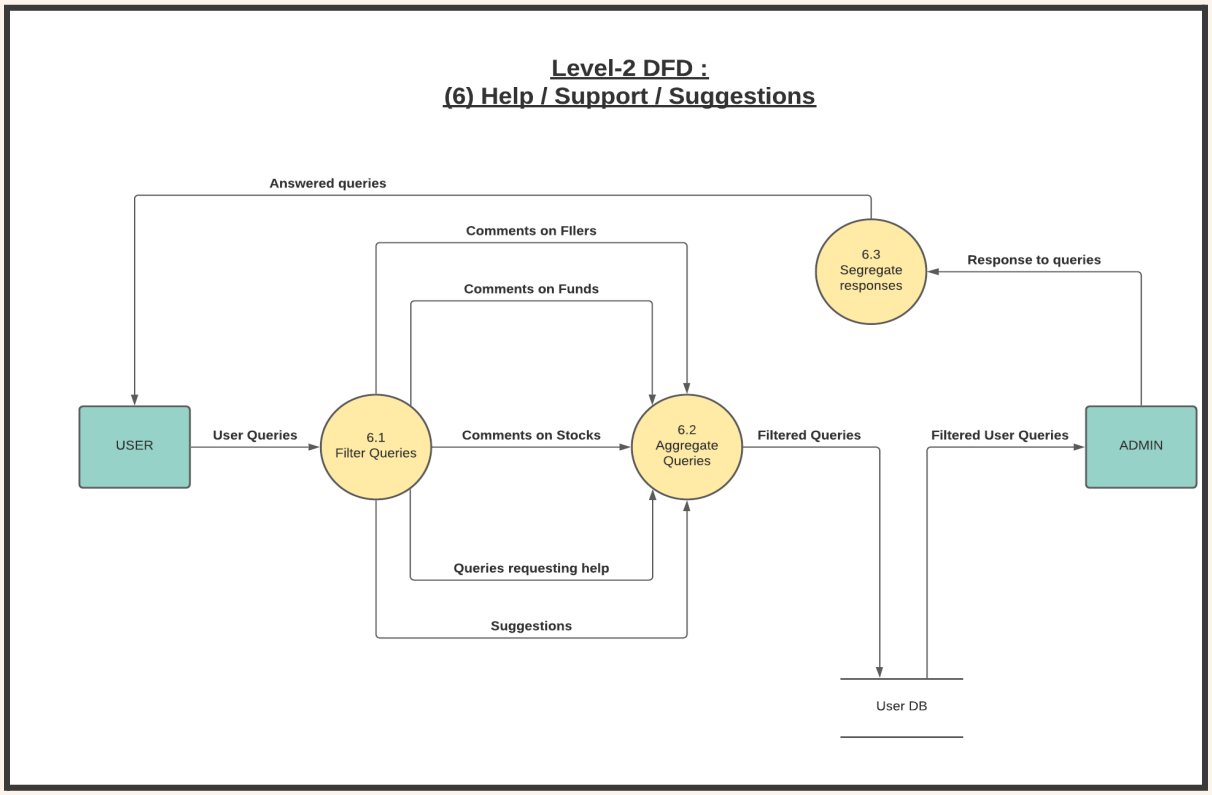
4.1.4 Level-2 DFD (4. Display Data, Charts & Tables)



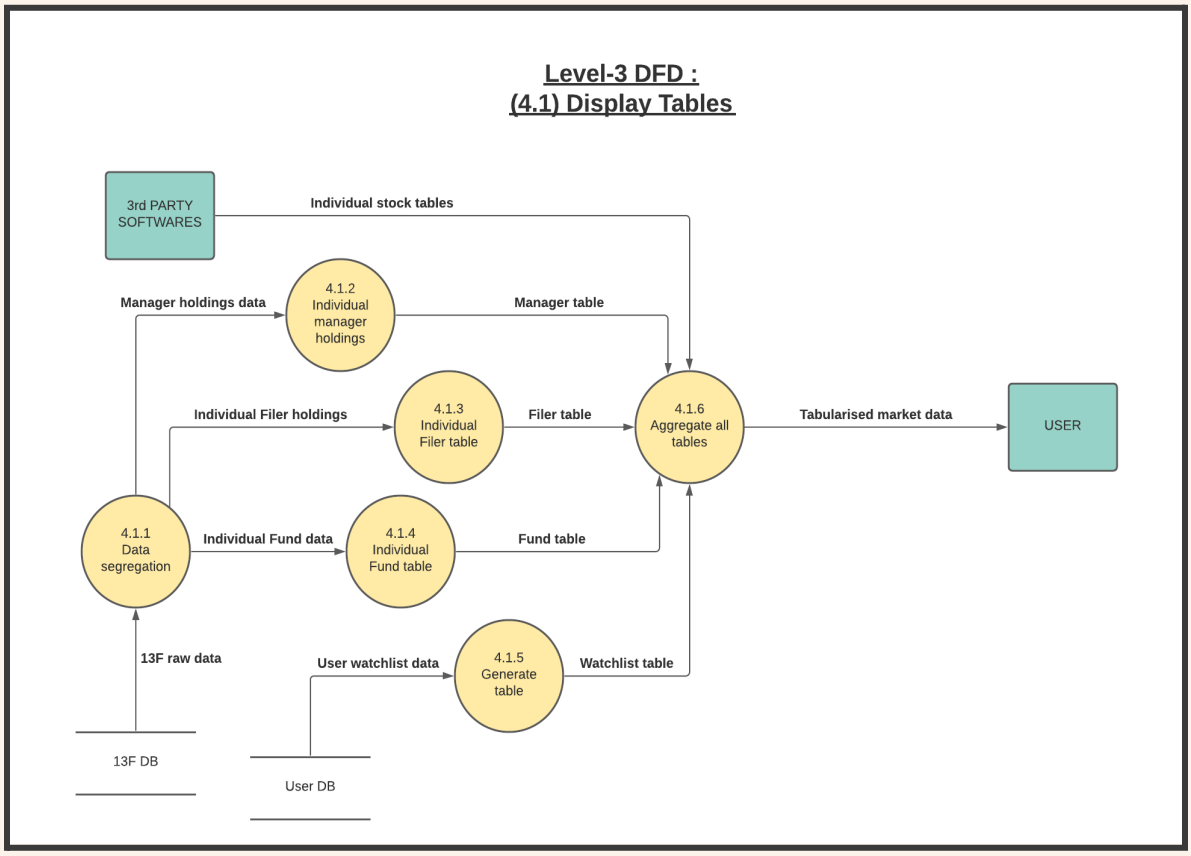
4.1.5 Level-2 DFD (5. Analyse Trends)



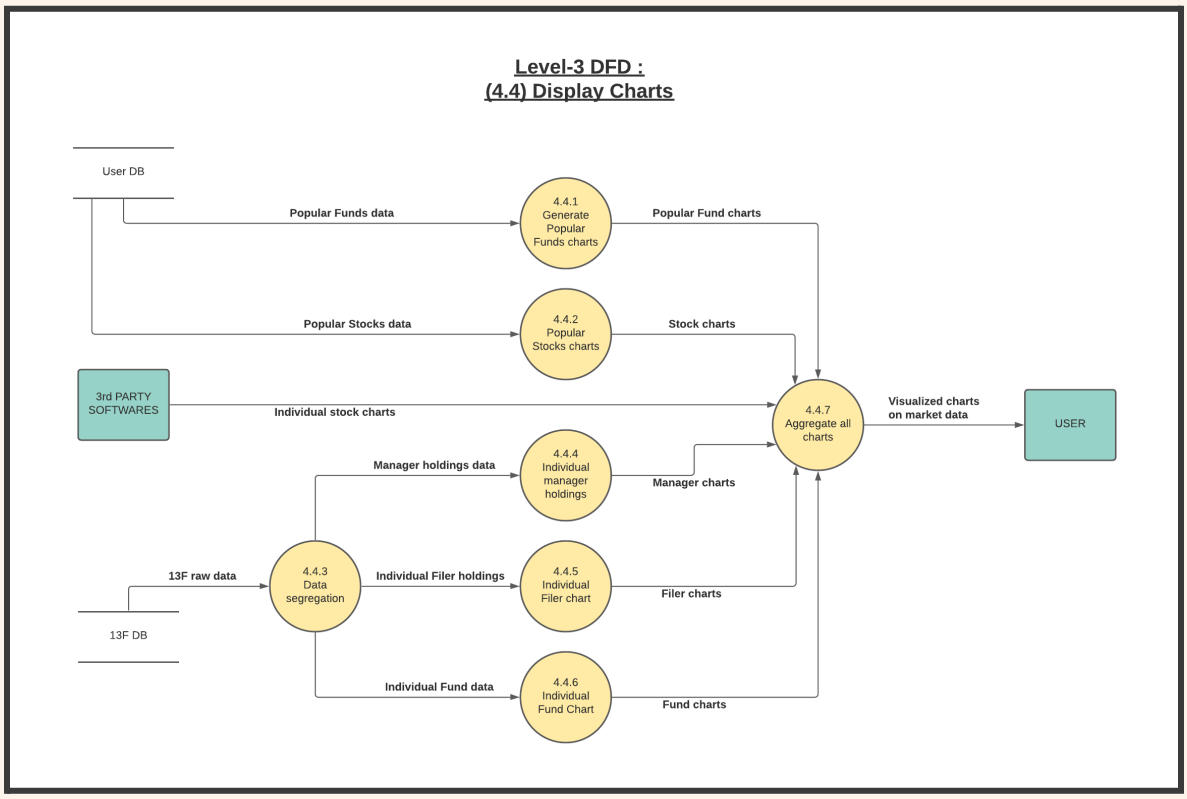
4.1.6 Level-2 DFD (6. Help/Support/Suggestions)



4.1.7 Level-3 DFD (4.1 Display Tables)



4.1.8 Level-3 DFD (4.4 Display Charts)



Use Cases

1. Sign Up

1.1 Brief description

This use case describes how the actors can Sign up in MrktDB.

1.2 Actor

The User and the admin are the actors in this use case.

1.3 Pre-condition

The actor should be present on the sign up page.

1.4 Flow of events

1.4.1 Basic flow

- The system will prompt the user to enter the Name, Email id, and Password.
- The actor will enter the above details asked by the system.
- A verification email will be sent to the provided Email id.
- The user will click on the verification link which is enclosed in the received email.
- The system will store the data into the user database.

1.4.2 Exception flow

- If in the basic flow, the actor enters an email that was already present in the database, an error stating the same should be displayed.
- If the actor does not click on the verification email within 30 minutes, the link will be disabled and the whole sign up process would have to be started from scratch.

1.5 Post-condition

A message stating the successful creation of an account should be displayed and the actor should be redirected to the Sign In page.

2. Sign In

2.1 Brief description

This use case describes how the actor can Sign In into MrktDB.

2.2 Actor

The User and the admin are the actors in this use case.

2.3 Pre-condition

The actor should be present on the sign in page.

2.4 Flow of events

2.4.1 Basic flow

- The system will prompt the user to enter the Email id, and Password.
- The actor will enter the above details asked by the system.
- Provided credentials would be checked in the database, and if present the actor should be signed in successfully.

2.4.2 Exception flow

- If in the basic flow, the actor enters an email or password that is not present in the user database, then an error stating the same should be displayed and the actor must not be signed in.

2.5 Post-condition

The actor should be redirected to the dashboard.

3. Search

3.1 Brief description

This use case allows the actor to search for any Hedge Funds, FIlers, or stocks and then displays the page for the relevant search results.

3.2 Actor

The User and the admin are the actors in this use case.

3.3 Pre-condition

The actor should be signed in.

3.4 Flow of events

3.4.1 Basic flow

- The actor can perform the search through the navbar or from the homepage.
- Any of the categories in which the search is to be performed can be chosen from the options.
- Entered characters will then be looked into the database for the chosen category and a live response giving the search results will be shown.
- The actor will then click on any of the returned results and be redirected to the particular page.

3.4.2 Exception flow

- If the entered characters do not match to any items in the database, then a message stating “No results found!” should be displayed.

3.5 Post-condition

The actor should be redirected to the particular page he/she was looking for.

4. User Dashboard

4.1 Brief description

This use case describes the user dashboard on which the watchlisted funds and stocks can be viewed.

4.2 Actor

The User is the actor in this use case.

4.3 Pre-condition

The actor should be signed in.

4.4 Flow of events

4.4.1 Basic flow

- The actor can view his/her watchlisted funds and stocks on the dashboard page.
- Charts related to the top fund/stock of the actors' choice will also be shown here.
- Bookmarked items by the actor will be displayed below the charts.
- Top 4 most popular stocks on MrktDB will be displayed below the bookmarked items.

4.4.2 Alternative flow

- If the actor is new to the website, then a detailed description of what the dashboard page will show will be displayed to the actor.

4.5 Post-condition

The actor can open charts and tables and look at the analysed data.

5. Individual Fund Dashboard

5.1 Brief description

When the actor will search a hedge fund's name, it will take the actor to the specific fund's dashboard. This dashboard format will stay consistent across 8000+ hedge funds that will be tracked. Historic data about the holdings by the hedge funds will be displayed.

5.2 Actor

The User and the admin are the actors in this use case.

5.3 Pre-condition

The actor should be signed in.

5.4 Flow of events

5.4.1 Basic flow

- The actor can view the filer dashboard after searching for the particular filer name.
- The filer dashboard will then be displayed for the particular filer.
- Hedge Fund name, description, and associated details like strategy, manager, location, etc will be displayed.
- A quick summary table with metrics like Market value, number of positions, number of new buys/sells will also be shown.
- A news feed related to the hedge fund and its holdings.

- There will be basic summary tables like
    - Top Holdings
    - Top Buys
    - Top Sells
- Basic visualizations like
    - Performance over time compared to S&P 500
    - Industry/sector Allocation
    - Small/Mid/Micro-Cap Allocation

5.4.2 Alternative flow

- The actor can also view the filer dashboard by clicking on a watchlisted filer on the user dashboard.

5.5 Post-condition

The actor should be able to watchlist the filer and download the data in csv format.

6. Single Stock Dashboard

6.1 Brief description

When the actor will search a stock's ticker/name, it will be redirected to the specific stock's dashboard. This dashboard format will stay consistent across 10,000+ stocks that will be tracked.

6.2 Actor

The User and the admin are the actors in this use case.

6.3 Pre-condition

The actor should be signed in.

6.4 Flow of events

6.4.1 Basic flow

- The actor can view the stock dashboard after searching for the particular stock name or ticker.
- The stock dashboard will then be displayed for the particular stock.
- Stock's name, description, and associated details like industry, location, etc. will be displayed.
- A quick summary table with metrics like Market cap, shares outstanding, average volume, bid-ask spread, P/E ratio, etc. will also be shown.

- Market Data visualization – Most likely embedded charts from other vendors like TradingView that show the time series of price, volume, etc.
- Aggregated 13F data table showing things like aggregated number of shares held across hedge funds, number of new buys/sells, stock's popularity rating, etc will be displayed.

6.4.2 Alternative flow

- The actor can also view the stock dashboard by clicking on a watchlisted stock on the user dashboard.

6.5 Post-condition

The actor should be able to watchlist the stock and download the data in csv format.

7. FAQ

7.1 Brief description

The FAQ functionality in the application is meant to resolve basic doubts of the users about the website.

7.2 Actor

The User and the admin are the actors in this use case.

7.3 Pre-condition

The actor should be on the faq page.

7.4 Flow of events

7.4.1 Basic flow

- The users view answers to some basic questions on the faq page.
- The admin will put up answers to general questions about the website in this section.

7.5 Post-condition

The actor should be able to ask questions to the admin in case the answer is not available on the faq page.

8. Contact Us

8.1 Brief description

Contact/Help function allows users to send an email to the admin in case of any queries/suggestions for the website.

8.2 Actor

The User and the admin are the actors in this use case.

8.3 Pre-condition

The actor should be on the contact us page and a message should be written.

8.4 Flow of events

8.4.1 Basic flow

- The users can ask questions to the admin about anything on the website.
- The admin can reply to the queries on the provided email id.

8.4.2 Exception flow

- If an email id is not provided by the user, then the admin cannot reply to the query.

8.5 Post-condition

If the admin feels that the question is really sensible, then it can be added to the FAQ page.

9. 404 PAGE

9.1 Brief description

If an actor tries to access any page which is not defined for the application, then it will redirect to a 404 page.

9.2 Actor

The User and the admin are the actors in this use case.

9.3 Pre-condition

The actor should be trying to search for a page on the application.

9.4 Flow of events

9.4.1 Basic flow

- The actor will try to search for a page on the application.
- If the page does not exist, then the application will redirect the actor to a 404 page stating “Page Not Found!”.

9.5 Post-condition

A button linking to the homepage can be pressed by the actor to return to the homepage.

10. Database Updation & Creation

10.1 Brief description

The database updation & creation script is to be run by the admin to create the database and update it every quarter.

10.2 Actor

The admin is the actor in this use case.

10.3 Pre-condition

The actor should have access to the database and the cloud server for running the scripts.

10.4 Flow of events

10.4.1 Basic flow

- The creation script will be run at the time when the application will be put out to general use.
- After that, an update script will run every quarter to look for new filings by the hedge funds.

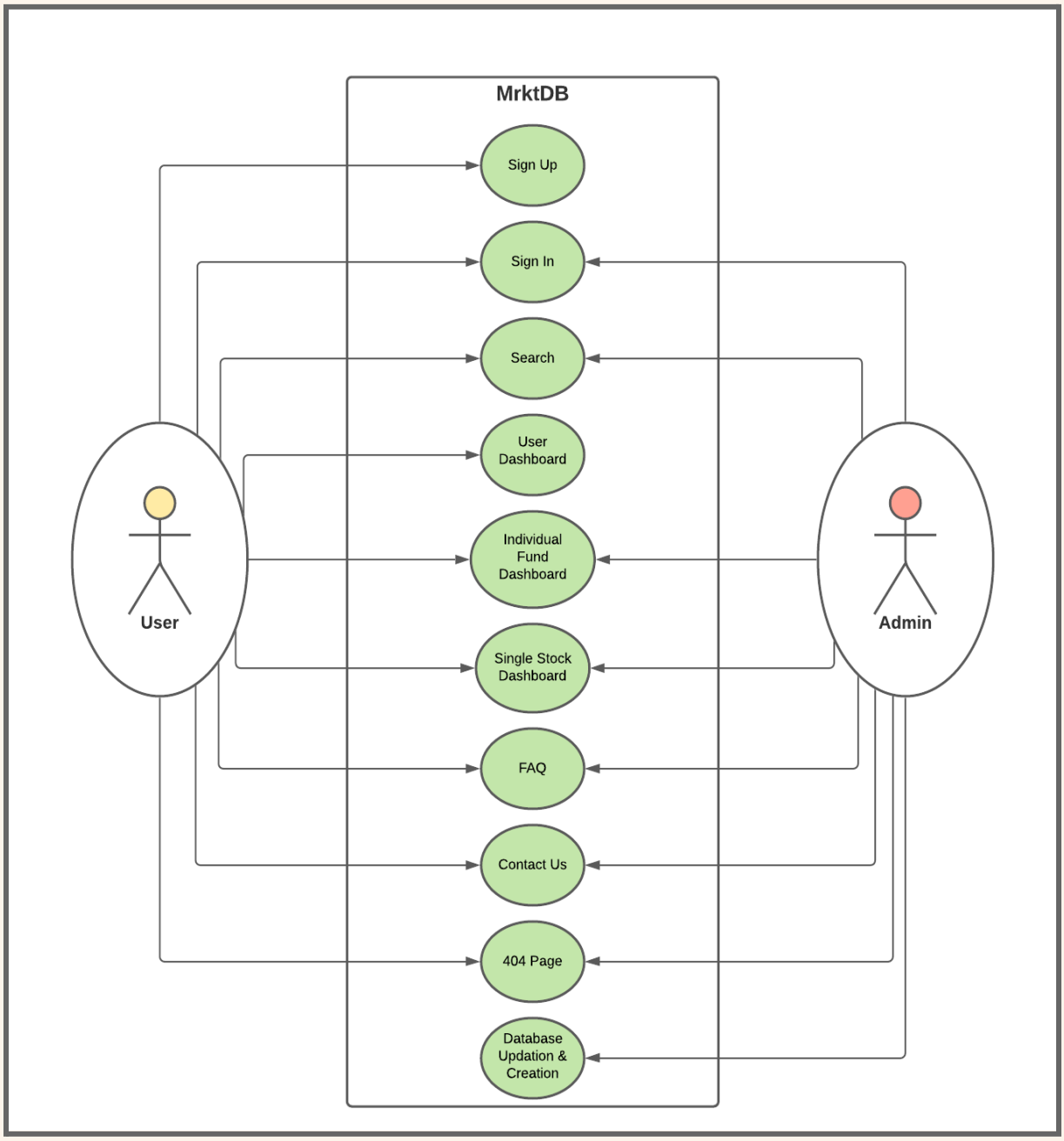
10.4.2 Exception flow

- If any problem occurs during database creation or updation then the script must be revised and the required changes need to be done.

10.5 Post-condition

The users and admin must then be able to access the new data through the application.

Use Case Diagram



Function Point Estimation

Computing Function Points

The FP metric can be used to-

(1) estimate the cost or effort required to design, code, and test the software;

(2) predict the number of errors that will be encountered during testing; and

(3) forecast the number of components and/or the number of projected source lines in the implemented system.

Function points are derived using an empirical relationship based on countable measures of software's information domain and qualitative assessments of software complexity. Information domain values are defined in the following manner :-

- **Number of external inputs (EIs):** Each *external input* originates from a user or is transmitted from another application and provides distinct application-oriented data or control information. Inputs are often used to update *internal logical files* (ILFs). Inputs should be distinguished from inquiries, which are counted separately.

- **Number of external outputs (EOs):** Each *external output* is derived data within the application that provides information to the user. In this context external output refers to reports, screens, error messages, and the like. Individual data items within a report are not counted separately.

- **Number of external inquiries (EQs):** An *external inquiry* is defined as an online input that results in the generation of some immediate software response in the form of an online output (often retrieved from an ILF).

- **Number of internal logical files (ILFs):** Each *internal logical file* is a logical grouping of data that resides within the application's boundary and is maintained via external inputs.

- **Number of external interface files (EIFs):** Each *external interface file* is a logical grouping of data that resides external to the application but provides data that may be of use to the application.

Information Domain Value	**Count**	**Weighting Factor**			**Total**
		Simple	**Average**	**Complex**	
External Inputs (EIs)	5	3	4	6	24
External Outputs (EOs)	49	4	5	7	307
External Inquiries (EQs)	8	3	4	6	44
Internal Logical Files (ILFs)	20	7	10	15	215
External Interface Files (EIFs)	2	5	7	10	20
Count total					**610**

$$\mathbf{FP = Count\ total * [0.65 + 0.01 * \Sigma(Fi)]}$$

, where count total is the sum of all FP entries obtained from the table above & *Fi* (*i* = 1 to 14) are *value adjustment factors* (VAF) based on responses to 14 predefined questions.

Value Adjustment Factor (VAF) or Σ(Fi)

S.No.	Questions	Grade Value
1.	Does the system require reliable backup and recovery?	**5**
2.	Are specialized data communications required to transfer information to or from the application?	**5**
3.	Are there distributed processing functions?	**5**
4.	Is performance critical?	**5**
5.	Will the system run in an existing, heavily utilized operational environment?	**4**
6..	Does the system require online data entry?	**5**
7.	Does the online data entry require the input transaction to be built over multiple screens or operations?	**5**
8.	Are the ILFs updated online?	**5**
9.	Are the inputs, outputs, files, or inquiries complex?	**5**
10.	Is the internal processing complex?	**5**
11.	Is the code designed to be reusable?	**4**
12.	Are conversion and installation included in the design?	**5**
13.	Is the system designed for multiple installations in different organizations?	**3**
14.	Is the application designed to facilitate change and ease of use by the user?	**5**
Σ(Fi)		**66**

VAF = Σ(*Fi*) = 66

Rate each question on a scale of 0 to 5:

0 : No influence

1 : Incidental

2 : Moderate

3 : Average

4 : Significant

5 : Essential

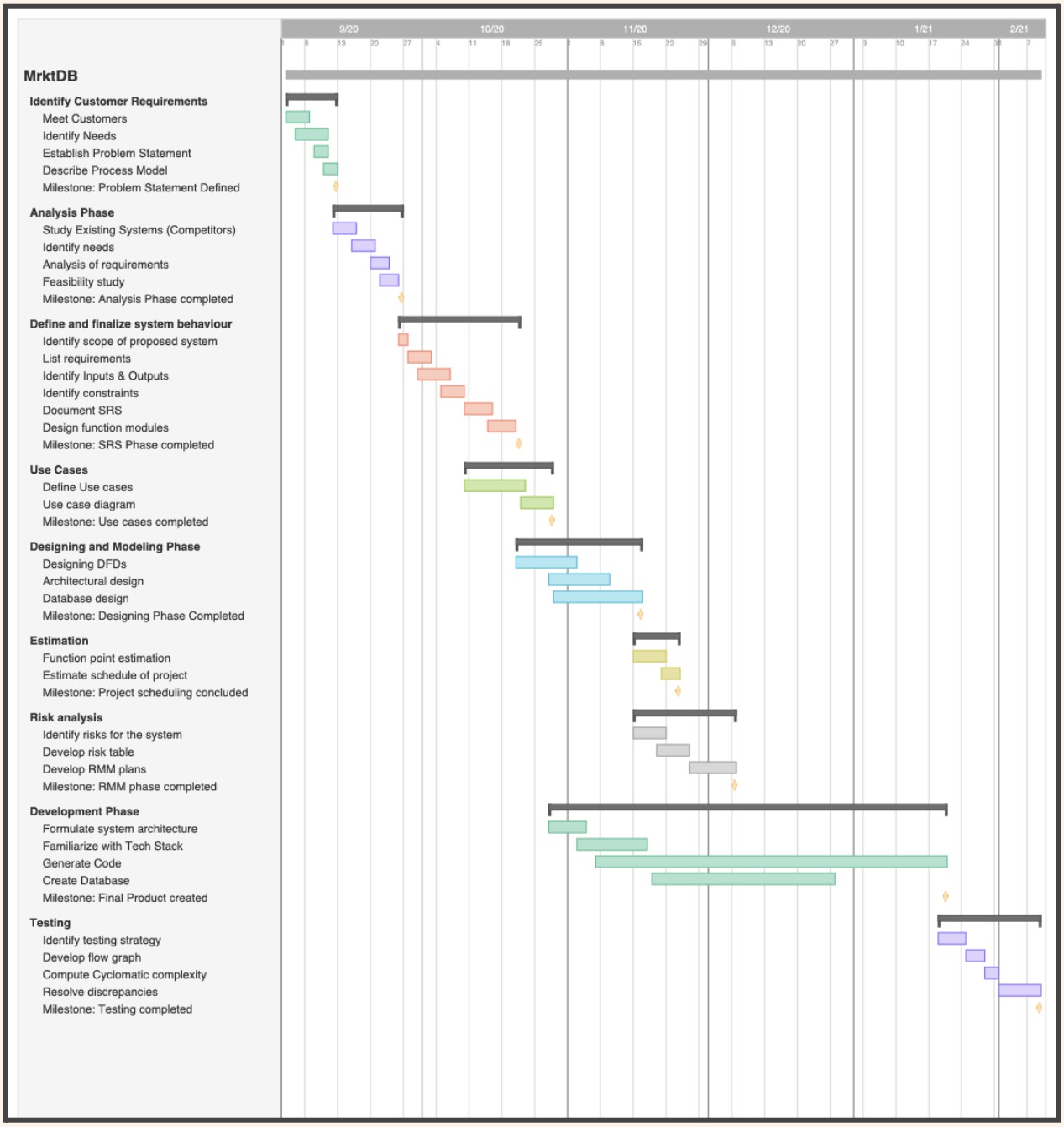
Function Point Metric = Count total * [0.65 + 0.01 * Σ(*Fi*)]

= 610 * [0.65 + 0.66] = 610 * 1.31

= 799.1

Scheduling

Timeline Chart



Risk Management

Risk Description	Probability of Occurrence	Impact Type	Loss Size (Days)	Risk Exposure (Days)
Project cost exceeds	50%	1	20	10
Technology doesn't meet expectations	30%	2	25	7.5
Lack of Promotion	40%	3	10	4
Miscommunication in team	40%	2	10	4
Data size estimation may be significantly low	60%	1	15	9
Privacy Issues	20%	2	5	1
Reputational Risk	20%	2	15	3
Insufficient time to validate on all browsers and OS types	45%	2	6	2.7
Following end user testing, more effort on user guide may be needed	25%	2	18	4.5
Problem in 3rd party solutions being used	25%	1	12	3

Risk Exposure (RE) = Probability * Loss size

Total Risk Exposure = Σ(RE) = 48.7

Impact values:

1. Catastrophic
2. Critical
3. Marginal
4. Negligible

Risk Mitigation, Monitoring and Management (RMMM)

1. Mitigation

In a financial website, reputational risk is probably the most critical risk that can occur due to the customer's inability to handle the software or if the system displays incorrect data that can result in a huge loss for the user. This would lead to decrease in revenues and increase in costs. So, special care has to be taken for proper functioning of the system.

The cost of the project would also rise too much if the requirements are changed after the initial steps have commenced. To mitigate this risk we would want to:

- Make sure the data collected is authentic and as extensive as possible.
- Put forward a deadline for proposing changes after which charges would be levied for each subsequent change proposed.

2. Monitoring

While working on the SRS, we should conduct multiple reviews to make sure the requirements are understood well and do not have to be changed much later.

Reputational risk can be monitored by releasing the product only after it is good enough for general use. Online mini tutorials can be displayed on the system that would enable users to work with the system more easily.

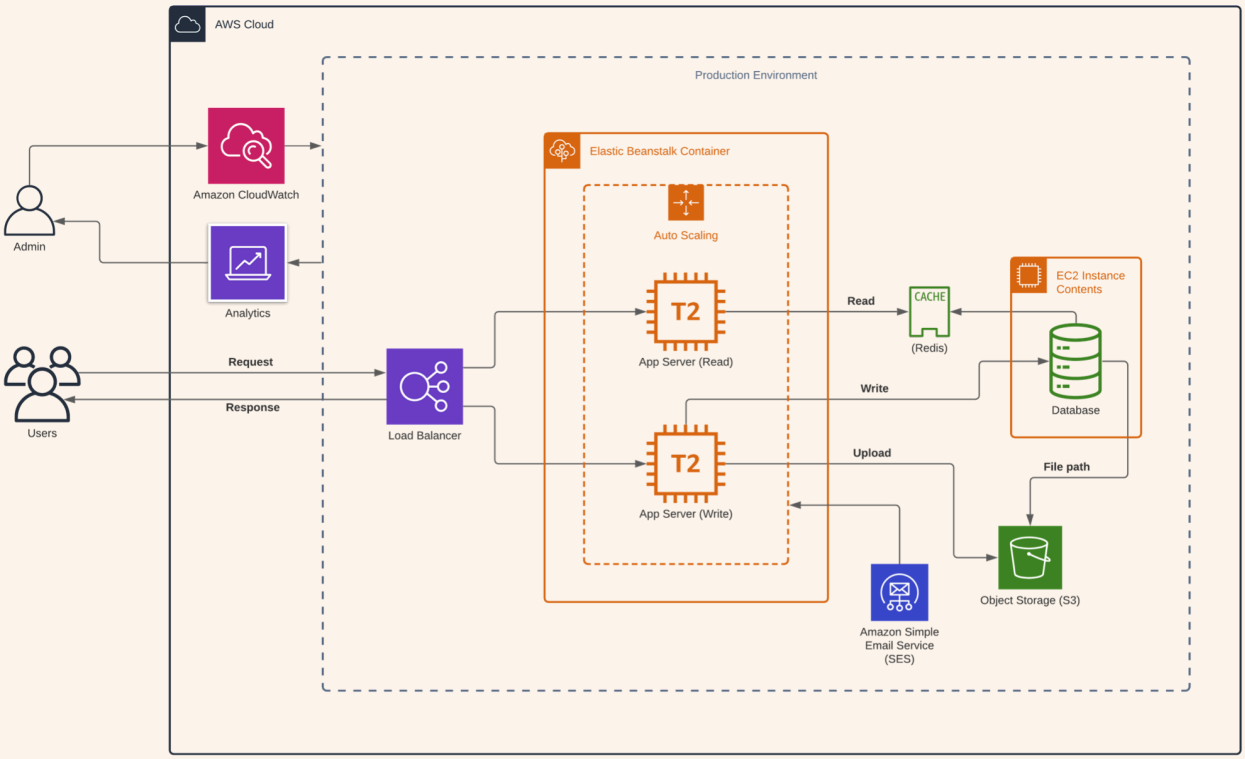
3. Management

In case there is no other option but to make a change in the SRS, the development team must cease their work until the change in the requirements is done and negotiated. This change must be flexible and thus the project could be continued without much hiatus.

Management of risks can further be done by establishing a healthy communicative environment within the employees. This would result in better team understanding and would also help in faster development.

Design

System Architecture Design



To view a better version, click here:
https://lucid.app/lucidchart/invitations/accept/inv_7d66b9a2-6939-4750-b178-fe000f173e9d

Database Design

Company			
Attribute	**Data Type**	**Size**	**Constraint**
Name	VARCHAR	80	NOT NULL
CIK	VARCHAR	10	NOT NULL
Company_Id	INT	10	Primary Key
Address	VARCHAR	150	NOT NULL
Company_Type	CHAR	2	NOT NULL

Filer			
Attribute	**Data Type**	**Size**	**Constraint**
Company_Id	INT	10	Foreign Key
File_Number	VARCHAR	10	NOT NULL
Filer_Id	INT	10	Primary Key
File_Type	CHAR	3	NOT NULL

QuarterlyOtherManager			
Attribute	**Data Type**	**Size**	**Constraint**
Parent_Filer_Id	INT	10	Foreign Key
Child_Filer_Id	INT	10	Foreign Key
Quarterly_Holding_Id	INT	10	Foreign Key
Number	INT	10	NOT NULL
Other_Manager_Id	INT	10	Primary Key

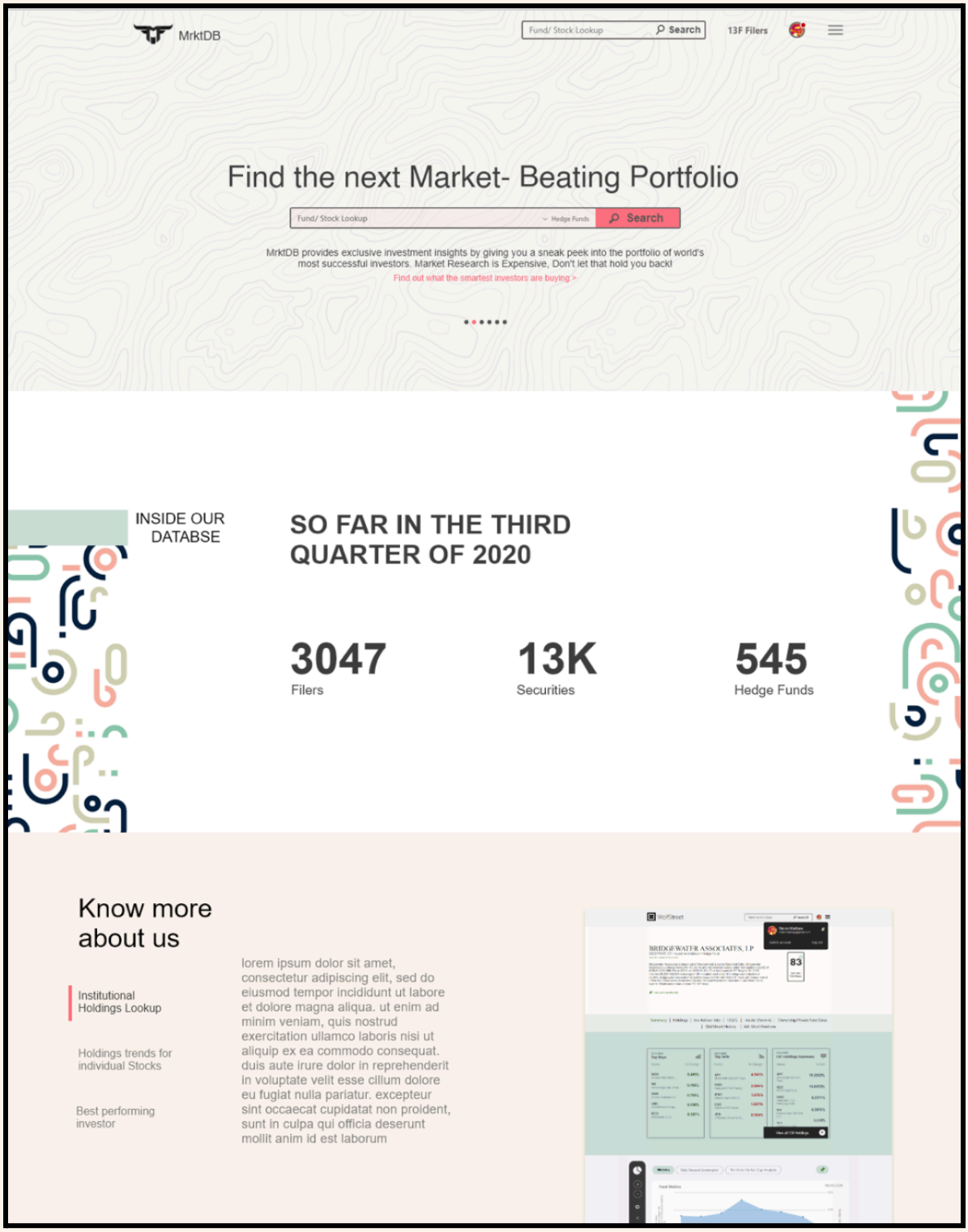
QuarterlyHolding			
Attribute	**Data Type**	**Size**	**Constraint**
Filer_Id	INT	10	Foreign Key
Quarter	INT	5	NOT NULL
Filing_Type	CHAR	3	NOT NULL
Filed_On	DATE	-	NOT NULL
Accepted_At	DATE	-	NOT NULL
Total_Value	INT	13	NOT NULL
Total_Entry	INT	5	NOT NULL
Quarterly_Holding_Id	INT	10	Primary Key

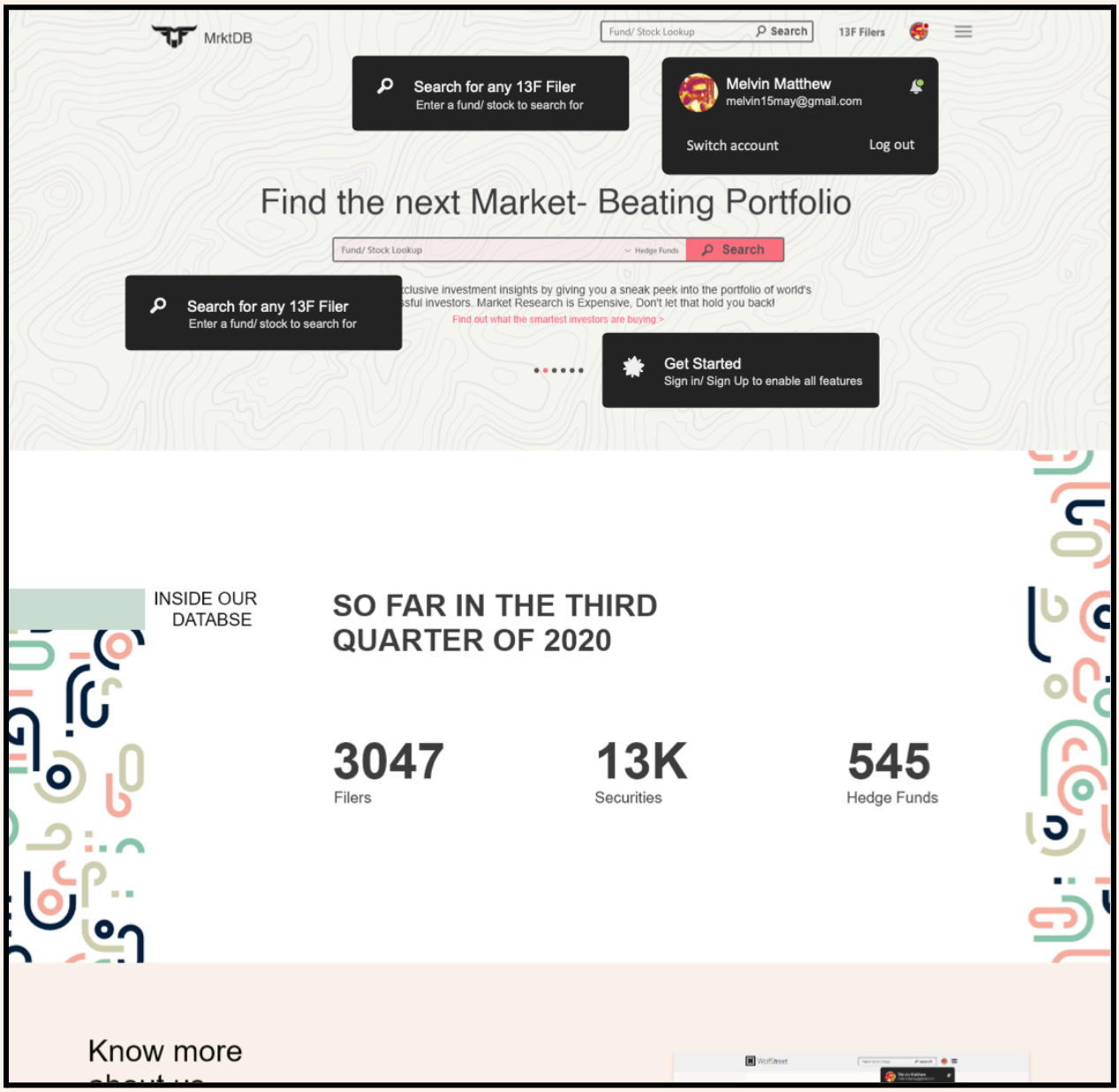
Security			
Attribute	**Data Type**	**Size**	**Constraint**
Company_Id	INT	10	Foreign Key
Cusip	INT	10	NOT NULL
Security_Name	INT	80	NOT NULL
Security_Type	CHAR	20	NOT NULL
Ticker	CHAR	5	NOT NULL
Title_Of_Class	INT	20	NOT NULL
Security_Id	INT	10	Primary Key

QuarterlySecurityHolding			
Attribute	**Data Type**	**Size**	**Constraint**
securityId	INT	10	Foreign Key
quarterlyHoldingId	INT	10	Foreign Key
marketvalue	INT	13	NOT NULL
quantity	INT	10	NOT NULL
holdingType	CHAR	4	NOT NULL
investmentDiscretion	CHAR	4	NOT NULL
sole	INT	10	NOT NULL
shared	INT	10	NOT NULL
none	INT	10	NOT NULL
quarterlySecurityHoldingId	INT	10	Primary Key

QuarterlyOtherManagerDistribution			
Attribute	**Data Type**	**Size**	**Constraint**
Quarterly_Other_Manager_Id	INT	10	Foreign Key
Quarterly_Security_Holding_Id	INT	10	Foreign Key
Quarterly_Other_Manager_Distribution_Id	INT	10	Primary Key

User Interface Design

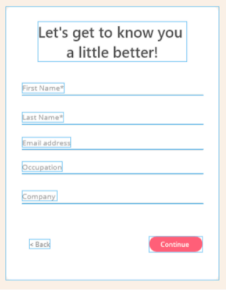



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Search
13F Filers
Search for any 13F Filer
Enter a fund/ stock to search for
Melvin Matthew
melvin15may@gmail.com
Switch account
Log out
Find the next Market- Beating Portfolio
Fund/ Stock Lookup
Hedge Funds
Search
Search for any 13F Filer
Enter a fund/ stock to search for
Find out what the smartest investors are buying.>
Get Started
Sign in/ Sign Up to enable all features
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SO FAR IN THE THIRD QUARTER OF 2020
3047
Filers
13K
Securities
545
Hedge Funds
Know more

LAYOUT OF SIGN UP PAGE



Sign up
Will ask user to sign up using email id or using Google or Facebook*



Will ask user to enter data

First & Last Name is compulsory. Email address already filled in previous page

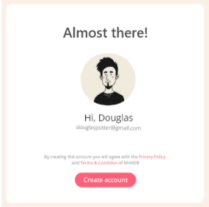
Back button available



Will ask user to upload an image

In case user uploads he's/ her image ask to crop and upload

Skip button available
No back button



Will show user entered detail

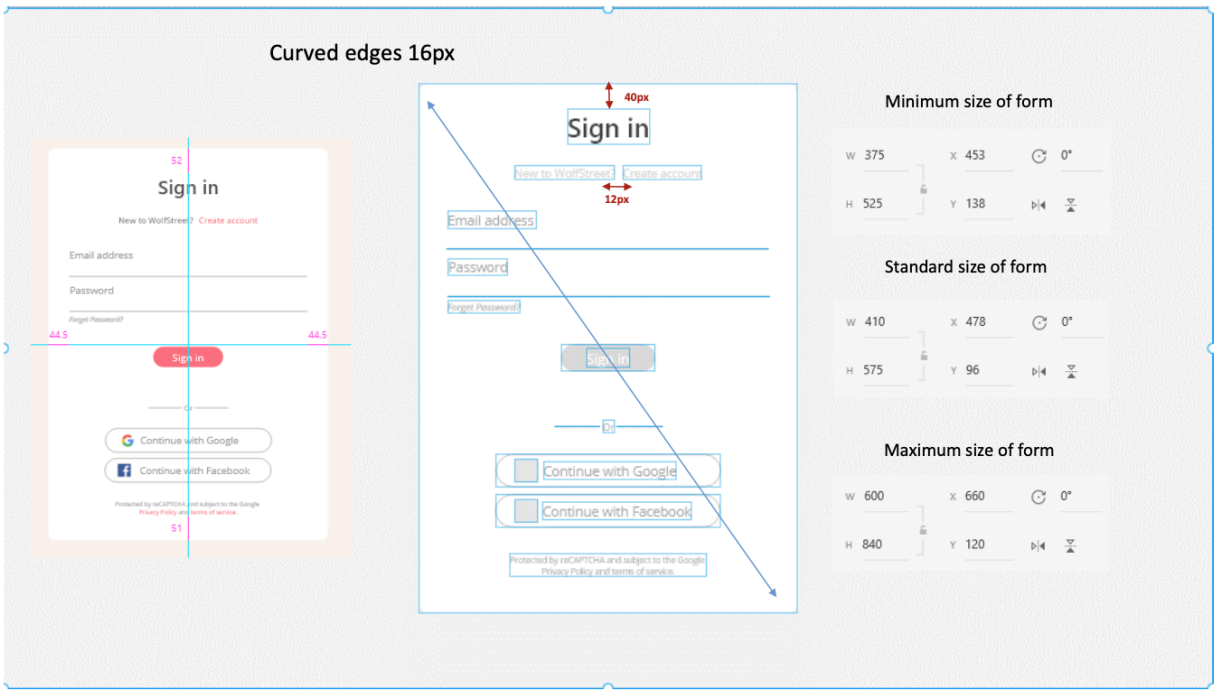
Incase user picture is available show user picture
Else show the first work of he's/her name

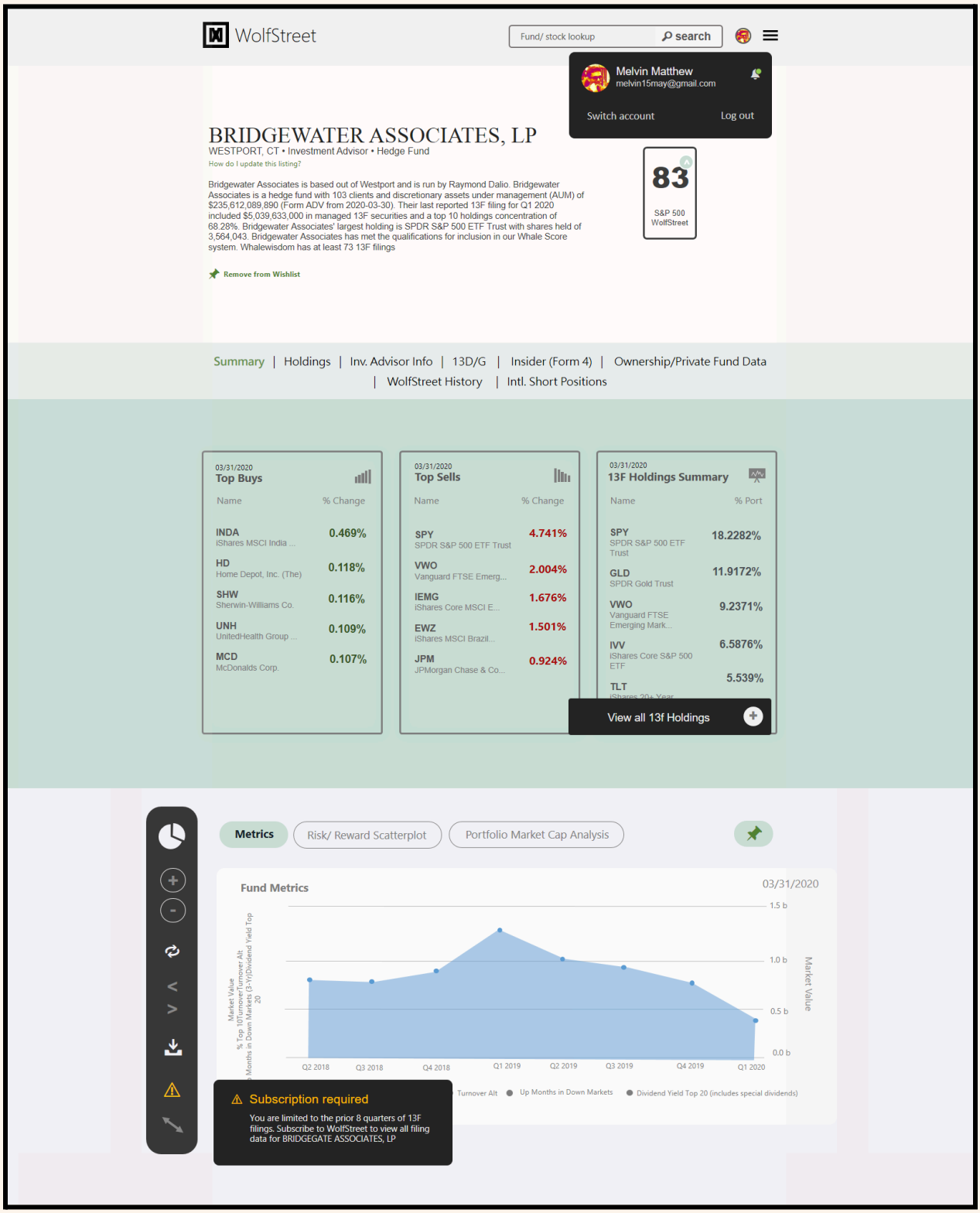
Example if user name is Douglas Potter and he didn't uploads he's profile picture the image shown will be the first word of his name 'DP'

No back or skip button available

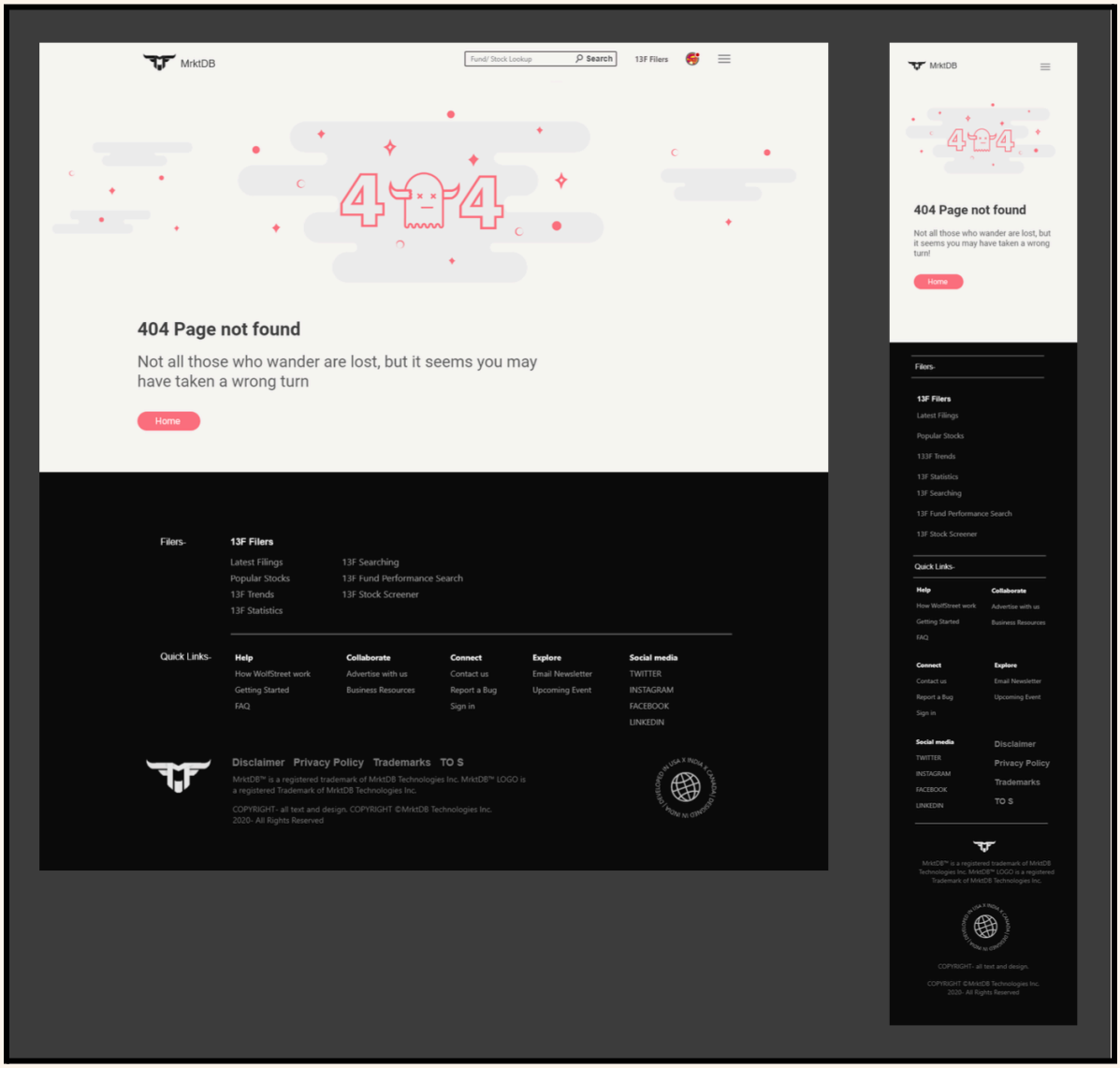


Example if user name is Douglas Potter and he didn't uploads he's profile picture the image shown will be the first word of his name 'DP' for Douglas Potter




WolfStreet
Fund/ stock lookup
search
Melvin Matthew
melvin15may@gmail.com
Switch account
Log out
BRIDGEWATER ASSOCIATES, LP
WESTPORT, CT • Investment Advisor • Hedge Fund
How do I update this listing?
Bridgewater Associates is based out of Westport and is run by Raymond Dalio. Bridgewater Associates is a hedge fund with 103 clients and discretionary assets under management (AUM) of $235,612,089,890 (Form ADV from 2020-03-30). Their last reported 13F filing for Q1 2020 included $5,039,633,000 in managed 13F securities and a top 10 holdings concentration of 68.28%. Bridgewater Associates' largest holding is SPDR S&P 500 ETF Trust with shares held of 3,564,043. Bridgewater Associates has met the qualifications for inclusion in our Whale Score system. Whalewisdom has at least 73 13F filings
Remove from Wishlist
83
S&P 500
WolfStreet
Summary | Holdings | Inv. Advisor Info | 13D/G | Insider (Form 4) | Ownership/Private Fund Data | WolfStreet History | Intl. Short Positions
03/31/2020
Top Buys
Name
% Change
INDA
iShares MSCI India ...
0.469%
HD
Home Depot, Inc. (The)
0.118%
SHW
Sherwin-Williams Co.
0.116%
UNH
UnitedHealth Group ...
0.109%
MCD
McDonalds Corp.
0.107%
03/31/2020
Top Sells
Name
% Change
SPY
SPDR S&P 500 ETF Trust
4.741%
VWO
Vanguard FTSE Emerg...
2.004%
IEMG
iShares Core MSCI E...
1.676%
EWZ
iShares MSCI Brazil...
1.501%
JPM
JPMorgan Chase & Co...
0.924%
03/31/2020
13F Holdings Summary
Name
% Port
SPY
SPDR S&P 500 ETF Trust
18.2282%
GLD
SPDR Gold Trust
11.9172%
VWO
Vanguard FTSE Emerging Mark...
9.2371%
IVV
iShares Core S&P 500 ETF
6.5876%
TLT
5.539%
View all 13f Holdings
Metrics
Risk/ Reward Scatterplot
Portfolio Market Cap Analysis
Fund Metrics
03/31/2020
1.5 b
1.0 b
0.5 b
0.0 b
Market Value
Q2 2018
Q3 2018
Q4 2018
Q1 2019
Q2 2019
Q3 2019
Q4 2019
Q1 2020
Turnover Alt
Up Months in Down Markets
Dividend Yield Top 20 (includes special dividends)
Subscription required
You are limited to the prior 8 quarters of 13F filings. Subscribe to WolfStreet to view all filing data for BRIDGEGATE ASSOCIATES, LP


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133F Trends
13F Statistics
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13F Fund Performance Search
13F Stock Screener
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Getting Started
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Collaborate
Advertise with us
Business Resources
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Report a Bug
Sign in
Explore
Email Newsletter
Upcoming Event
Social media
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INSTAGRAM
FACEBOOK
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Report a Bug
Sign in
Collaborate
Advertise With Us
Business Resources
Explore
Email Newsletter
Upcoming Event
HAMBURGER MENU
FIG 1
FIG 2
FIG 1
The footer section will cover full VW (view width) horizontally and to its full length vertically until its contents are over.
FIG 2
The menu section will cover full VW & VH


MrktDB
Fund/ Stock Lookup
Search
13F Filers
404
404 Page not found
Not all those who wander are lost, but it seems you may have taken a wrong turn
Home
Filers-
13F Filers
Latest Filings
Popular Stocks
13F Trends
13F Statistics
13F Searching
13F Fund Performance Search
13F Stock Screener
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Advertise with us
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Sign in
Explore
Email Newsletter
Upcoming Event
Social media
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Programming/Coding

```python
import requests
import csv
from io import BytesIO
from zipfile import ZipFile
import urllib.request
from bs4 import BeautifulSoup

def append_data(file_name, list_of_elem):
    with open(file_name, 'a+', newline='') as write_obj:
        writer = csv.writer(write_obj)
        writer.writerows(list_of_elem)

def download_csv():
    DOMAIN = "https://www.sec.gov"
    FAILS_TO_DELIVER_URL = "https://www.sec.gov/data/foiadocsfailsdatahtm"
    req = requests.get(FAILS_TO_DELIVER_URL)
    soup = BeautifulSoup(req.content, 'html.parser')

    a_list = soup.find_all('a')

    for a_href in a_list:
        if a_href.startswith("/files/data/"):
            file_url = DOMAIN + a_href
            url = urllib.request.urlopen(file_url)

            with ZipFile(BytesIO(url.read())) as my_zip_file:
                for contained_file in my_zip_file.namelist():
                    if contained_file == "cnsfails201212b.txt":
                        sys.exit(-1)

                    temp_file_name = contained_file.split('.')
                    file_name = temp_file_name[0] + ".csv"
                    print(file_name)
                    data = []
```

```
                    for line in my_zip_file.open(contained_file).readlines():
                        line=line.decode('ISO-8859-1')
                        row = line.split('|')
                        data.append(row)

                    del data[-1]
                    del data[-1]

                    append_data(file_name, data)

            elif a_href.startswith("/files/node/add/"):
                file_url = DOMAIN + a_href
                ftd_txt = urllib.request.urlopen(file_url)

                if a_href.endswith("cnsfails201212b.txt"):
                    sys.exit(-2)

                temp_file_name = a_href.split('.')
                file_name = temp_file_name[0] + ".csv"
                print(file_name)
                data = []

                for line in ftd_txt.readlines():
                    line = line.decode('ISO-8859-1')
                    row = line.split('|')
                    data.append(row)

                del data[-1]
                del data[-1]

                append_data(file_name,data)

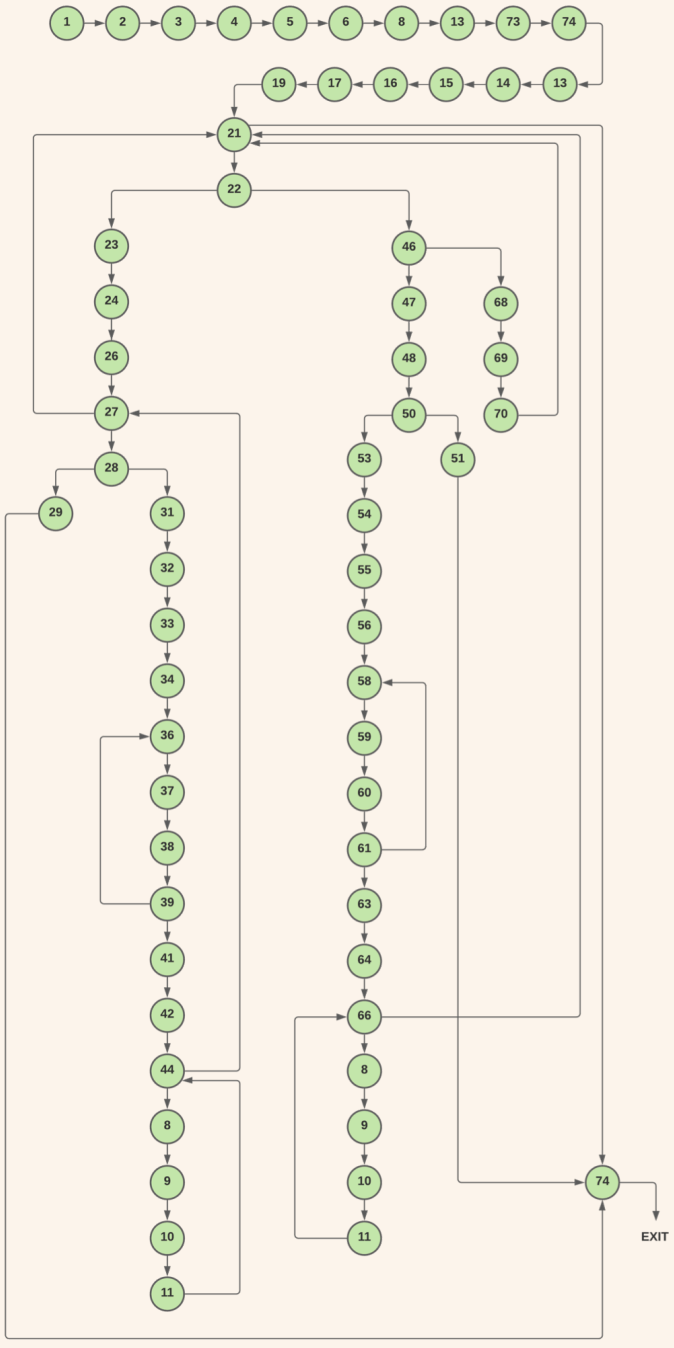
```

```
            else:
                print('url pattern did not match')
                continue


if __name__ == '__main__':
    download_csv()
```

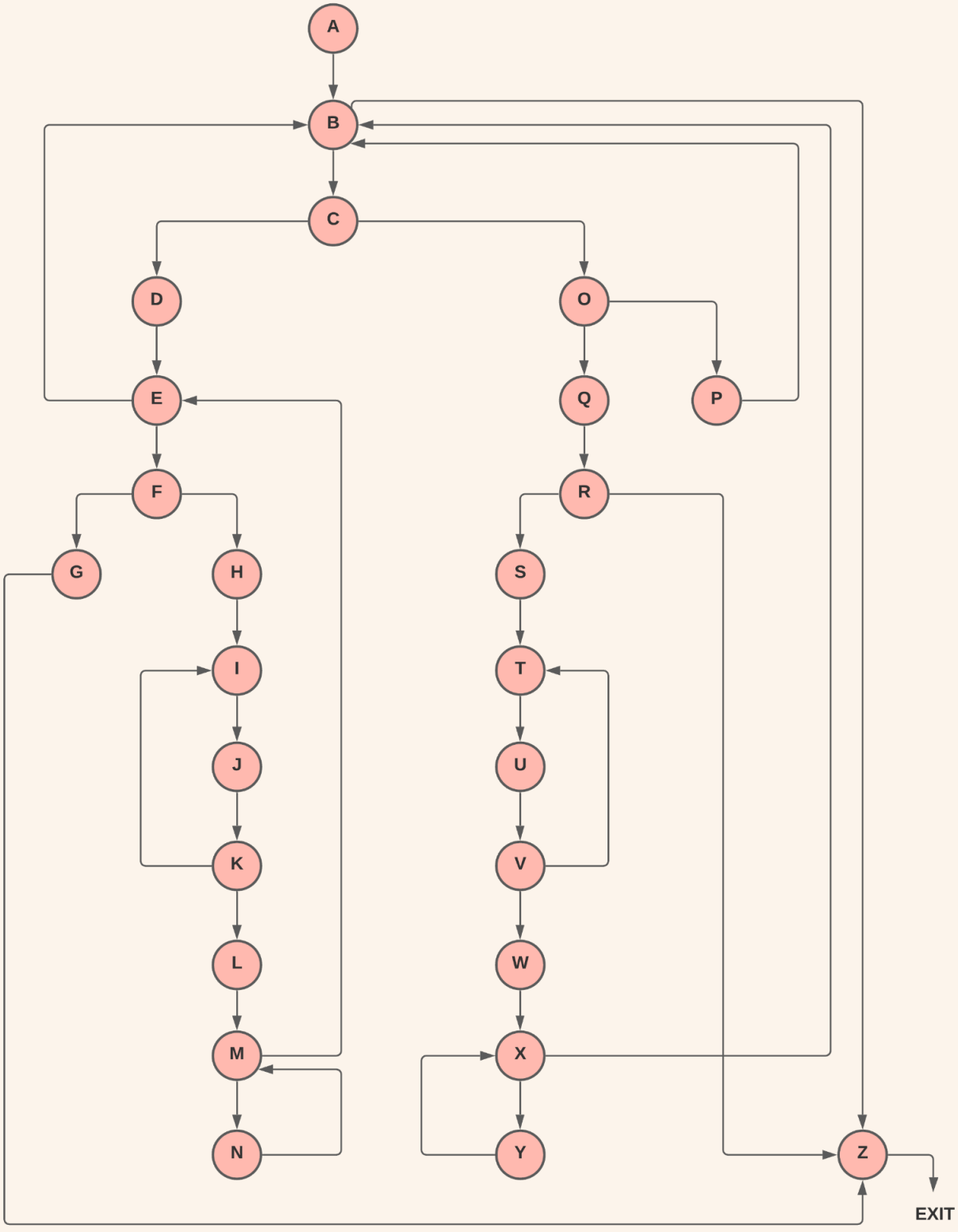
Testing

Path Testing (flow graph)



Decision-to-Decision(DD) Path Graph

Mapping Table		
Flow Graph Nodes	**DD Path Graph corresponding nodes**	**Remarks**
1, 2, 3, 4, 5, 6, 8, 13, 73, 74, 13, 14, 15, 16, 17, 19	A	Sequential nodes
21	B	Decision node
22	C	Decision node
23, 24, 26	D	Sequential nodes
27	E	Loop start node
28	F	Decision node
29	G	Exit node
31, 32, 33, 34	H	Sequential nodes
36	I	Loop start node
37, 38	J	Sequential nodes
39	K	Loop end node
41, 42	L	Sequential nodes
44	M	Loop end node
8, 9, 10, 11	N	Sequential nodes
46	O	Decision node
68, 69, 70	P	Sequential nodes
47, 48	Q	Sequential nodes
50	R	Decision node
53, 54, 55, 56	S	Sequential nodes
58	T	Loop start node
59, 60	U	Sequential nodes
61	V	Loop end node
63, 64	W	Sequential nodes
66	X	Loop end node
8, 9, 10, 11	Y	Sequential nodes
74	Z	Exit node


A
B
C
D
O
E
Q
P
F
R
G
H
S
I
T
J
U
K
V
L
W
M
X
N
Y
Z
EXIT

Independent paths-

1. ABCDEFGZ
2. ABCDEFHIJKLMEBZ
3. ABCDEFHIJKIJKLMEBZ
4. ABCDEFHIJKLMNMEBZ
5. ABCDEFHIJKIJKLMNMEBZ
6. ABCOPBZ
7. ABCOQRZ
8. ABCOQRSTUVWXBZ
9. ABCOQRSTUVTUVWXBZ
10. ABCOQRSTUVWXYXBZ
11. ABCOQRSTUVTUVWXYXBZ

Cyclomatic Complexity

1. McCabe's cyclomatic metric, V(G) = e - n + 2P

   , where e = no. of edges, n = no. of vertices, and P connected components.

   V(G) = 35 - 26 + 2*1 = 11

2. V(G) = **Π** + 1

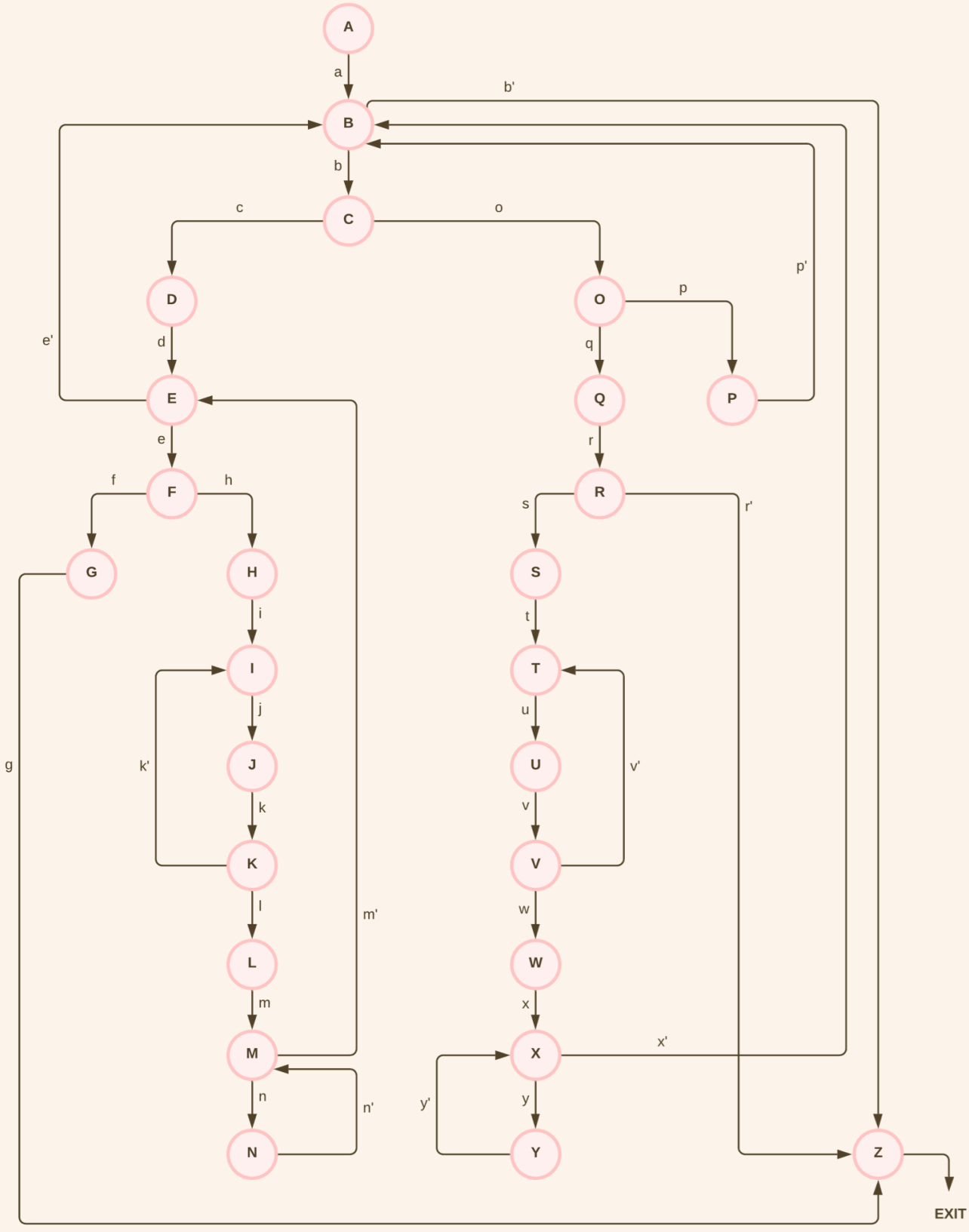
   , where **Π** is the number of predicate nodes contained in the flow graph G.

   ∴ V(G) = 10 + 1 = 11 (*Predicate nodes:* B, C, E, F, K, M, O, R, V, X)

3. Cyclomatic complexity is equal to the number of regions in the graphs = 11

Graph Matrix

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A		a																								
B			b																							b’
C				c											o											
D					d																					
E		e’				e																				
F							f	h																		
G																										g
H									i																	
I										j																
J											k															
K									k’			l														
L													m													
M					m’									n												
N													n’													
O																p	q									
P		p’																								
Q																		r								
R																			s							r’
S																				t						
T																					u					
U																						v				
V																				v’			w			
W																								x		
X		x’																							y	
Y																								y’		
Z																										


A
a
B
b'
b
C
c
o
D
O
p
p'
e'
d
q
E
Q
P
e
r
f
F
h
R
s
r'
G
H
S
i
t
I
T
j
u
g
k'
J
U
v'
k
v
K
V
l
w
m'
L
W
m
x
M
X
x'
n
n'
y'
y
N
Y
Z
EXIT

Connection Matrix

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	Connections
A		1																									1-1=0
B			1																							1	2-1=1
C				1											1												2-1=1
D					1																						1-1=0
E		1				1																					2-1=1
F							1	1																			2-1=1
G																										1	1-1=0
H									1																		1-1=0
I										1																	1-1=0
J											1																1-1=0
K									1			1															2-1=1
L													1														1-1=0
M					1									1													2-1=1
N													1														1-1=0
O																1	1										2-1=1
P		1																									1-1=0
Q																		1									1-1=0
R																			1							1	2-1=1
S																				1							1-1=0
T																					1						1-1=0
U																						1					1-1=0
V																				1			1				2-1=1
W																								1			1-1=0
X		1																							1		2-1=1
Y																								1			1-1=0
Z																											0
Total																											**10+1=11**

Boundary Value Analysis

```
filed_month = d2['reportCalendarOrQuarter'][:2]
year = d2['year']

if (filed_month == '03'):
    quarter = '1'
elif (filed_month == '06'):
    quarter = '2'
elif (filed_month == '09'):
    quarter = '3'
else:
    quarter = '4'

year_quarter = year + quarter
```

year_quarter = year + quarter

The value of the variable **quarter** ranges from [1, 4] and the value for the variable **year** is supposed to be in the range [2013, 2020].

Test cases: 4*n + 1 = 4*2 + 1 = 8 + 1 = 9

Min. year = 2013	Min. quarter = 1
Close to min. = 2014	Close to min. = 2
Nominal = 2017	Nominal = 3
Close to max. = 2019	Close to max. = 3
Max. year = 2020	Max. quarter = 4

Test Case	year	quarter	year_quarter	Validity
1	2017	1	20171	Valid
2	2017	2	20172	Valid
3	2017	3	20173	Valid
4	2017	4	20174	Valid
5	2017	3	20173	Valid
6	2013	3	20133	Valid
7	2014	3	20143	Valid
8	2019	3	20193	Valid
9	2020	3	20203	Valid

Robustness Testing

Lesser than min. year = 2012

Larger than max. year = 2021

Lesser than min. Quarter = 0

Larger than max. Quarter = 5

Total test cases = 6*n + 1 = 6*2 + 1 = 12 + 1 = 13. So, there will be 4 more test cases apart from the above 9 cases. They are:-

Test Case	year	quarter	year_quarter	Validity
1	2017	0	20170	Invalid
2	2017	5	20175	Invalid
3	2012	3	20123	Invalid
4	2021	3	20213	Invalid

These values are invalid because the result is either not possible or out of the domain as per the requirements.

Worst-case Testing

Here we reject the single fault assumption theory, and consider cases where more than 1 variable has extreme values.

Total test cases = $5^n = 5^2 = 25$.

The test cases will be a cross-product of the sets {2013, 2014, 2017, 2019, 2020} X {1, 2, 3, 4, 5}. They are:-

Test Case	year	quarter	year_quarter	Validity
1	2013	1	20131	Valid
2	2013	2	20132	Valid
3	2013	3	20133	Valid
4	2013	4	20134	Valid
5	2013	5	20135	Invalid
6	2014	1	20141	Valid
7	2014	2	20142	Valid
8	2014	3	20143	Valid
9	2014	4	20144	Valid
10	2014	5	20145	Invalid
11	2017	1	20171	Valid
12	2017	2	20172	Valid
13	2017	3	20173	Valid
14	2017	4	20174	Valid
15	2017	5	20175	Invalid
16	2019	1	20191	Valid
17	2019	2	20192	Valid
18	2019	3	20193	Valid
19	2019	4	20194	Valid
20	2019	5	20195	Invalid
21	2020	1	20201	Valid
22	2020	2	20202	Valid
23	2020	3	20203	Valid
24	2020	4	20204	Valid
25	2020	5	20205	Invalid

References

I. Pressman, R. S., & Maxim, B. R. (2015). Software Engineering: A Practitioner's Approach. 8th edition. McGraw-Hill.

II. Aggarwal, K. K., & Singh, Y. (2007). Software Engineering. 3rd edition. New Age International Publishers.

III. SEC Edgar: https://www.sec.gov/edgar/searchedgar/companysearch.html